

Kaggle Dataset: Uber vs. Lyft RideShare, Boston MA
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Statistical Analysis Project
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Abstract

This project analyses the comprehensive statistical analysis and predictive modeling of rideshare data, focusing on Uber and Lyft services in Boston, MA, using a Kaggle dataset consisting of 637,976 records and 57 features spanning from November 26, 2018, to December 18, 2018. The objectives are to predict ride pricing using regression analysis and to classify cab type using logistic regression and Naive Bayes models. Data preprocessing involved extensive cleaning and transformation to handle missing values, outliers, and multicollinearity, resulting in reducing the dataset to 41 key features. Exploratory Data Analysis (EDA) revealed significant patterns in pricing, surge multipliers, and weather-related variables, underscoring key correlations between price, distance, surge multiplier, and temperature. Box plots, scatter plots, and correlation matrices were used to visualize data distributions and highlight potential predictors. In the model-building phase, multiple linear regression was implemented to predict ride pricing. However, the model exhibited multicollinearity and non-linearity, causing further transformations and feature selection to refine its reliability. For cab type classification, both logistic regression and Naive Bayes models were developed, incorporating predictors such as surge multiplier, distance, and time-related features to enhance classification accuracy. Model validation was conducted using a 70/30 train-test split, evaluating model accuracy through key metrics such as MSE for regression and accuracy for classification. The regression model showed strong predictive capability, while classification models successfully differentiated between Uber and Lyft rides. The regression model demonstrated substantial predictive performances, but residual analysis flagged potential overfitting and variance inconsistency. In classification, Naive Bayes slightly outperformed logistic regression, effectively differentiating between Uber and Lyft rides. Overall, the project highlights the significance of data preprocessing, feature selection, and model validation in building reliable predictive models for rideshare data.

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Section 1: Dataset Information

Section 1.1: Introduction

The popularity of rideshare applications has skyrocketed in recent years as society moves towards more convenient and accessible modes of transportation and general life tendencies. On the forefront, Uber and Lyft, set the standard of rideshare applications, grossing around \$5-10 billion in annual revenue, with millions of active daily users at any moment. From the perspective of a data scientist, commercial service, social media or general quality-of-life applications provide some of the most valuable, sought-after data as they provide a deep and rich dimension often with an immensely large number of factors, transforming the problem domain into one with great value, while also fostering great tediousness. Fundamentally deconstructing the data into understandable patterns, or learning models is an immensely complex task; however, advertisers, commercial service providers, and even the clients themselves better understand, for example, what the exact factors that dictate the cost of a fare, or the factors that may lead to a particular selection of specialized cars. Such patterns are difficult to derive purely through insight; as such, we use powerful tools such as machine learning, or statistics such as regression and classification, to gap our understanding between the data and its outcome.

Section 1.2: Task Overview

With the following dataset, we aim to tackle two of its most defining features: rideshare, and Uber vs. Lyft. With Multiple Linear Regression, the goal is to develop a model that will allow accurate prediction of the response variable “price”, given a set of features. Moreover, we would also like to be able to see how classifiable are the two companies based on their customer preferences and subtle intrinsic and extrinsic factors. For this task, we will use two methods: Naive Bayes for the baseline, and then Logistical Regression to answer the objective of the cab type. The exploratory data analysis will be preliminarily conducted to provide insight into data, data trends, potential hidden patterns, or pitfalls.

Section 1.3: Data Overview

On Kaggle, “Uber and Lyft Dataset Boston, MA”, is the dataset selected to tackle the problem of prediction through regression and classification. The dataset features around 637,976 records, with 57 features, approaching approximately a file size of 367.38 MB. The dataset describes relevant data corresponding to a particular fare. The data was specifically taken from Boston, MA, from the date of 11-26-2018 to 12-18-2018, comprising roughly 2 months of data. There are two rideshare service providers featured in the dataset: Uber and Lyft. The dataset itself is very intricate, featuring obvious predictors such as distance, however, what makes the data even more interesting, is the unconventional or non-obvious features such as those relating to temperature, apparent temperature, non-temperature-related weather, and environmental factors. Furthermore, the dataset seems to have access to features that are possibly only accessible at higher levels of administration, such as *surge multiplier*, often used to dictate price based on statistical data of client demand. The overview of the dataset features is available in **Appendix A**. The data has been classified for both *price* as a response

variable, and *cab_type*, using various methods such as Decision Trees, Regression, RandomForest, Neural Networks, and so on.

Section 1.4: Data Preprocessing

Summary and Data Description: Summary statistics and measures of central tendency were used to examine and understand the distribution of the data. These included Mean, Mode, Standard Deviation, Min, Max, Quantile 1, and Quantile 2.

Missing Data and Invalid Data: A table was created with the number of missing data in each feature only price has NA values (55095, 7.95% of price values). Given the size of the dataset, the records with NA values were discarded. Additional analysis was performed to find out the data invalid like NaN and infinite values, but there were none. The final record count was 637,976.

Pre-Analysis Feature Dropping: Several features were considered either out of scope or irrelevant to our objective for the project. For example, time features included time in Unix Timestamp format which includes time by second, thus hindering the usefulness of these features in uncovering larger patterns. Moreover, some features were either redundant (i.e. duplicate columns like visibility and visibility.1), incomplete (i.e. product_id, which had a mix of unidentifiable ID and names), or would cause overfitting and provide no meaningful data (i.e. unique ID). This pre-analysis helped with dimensionality later during the model-building phase, but also helped understand the logistical depth of the dataset. The final feature count was 41.

Uniform Stratified Sampling: The sampling procedure is a significant aspect of our data pre-processing procedure. Essentially, we mutated three features, hour, day, and month, into a single categorical variable, and then performed sampling such that it extracts a random sample of X records from each “bin”, which is the hour-day-month. In other words, we want to extract X records for every hour of every day of every month that is in our dataset. For the most part, this had no problems, as only 4 days failed to meet our X limit, as such, we did not drop them, but let X be all the data they have. The significance of this process was to ensure that we would get an equal amount of data from each day, and prevent biased data sampling that may be fixated on a particular time, day or month, preventing us from performing proper data analysis

Section 1.5: Organization of Project

The report is segmented into four main sections: Firstly, we will perform exploratory data analysis which consists of both descriptive and inferential statistics. Secondly, we will begin building a model for regression (multivariate linear regression), and classification (Naive Bayes, decision trees, and logistical regression). Thirdly, the models will be validated through appropriate metrics and statistical structures. Lastly, we will finalize our findings in the form of a discussion & conclusion section, highlighting our findings, main challenges, and future works.

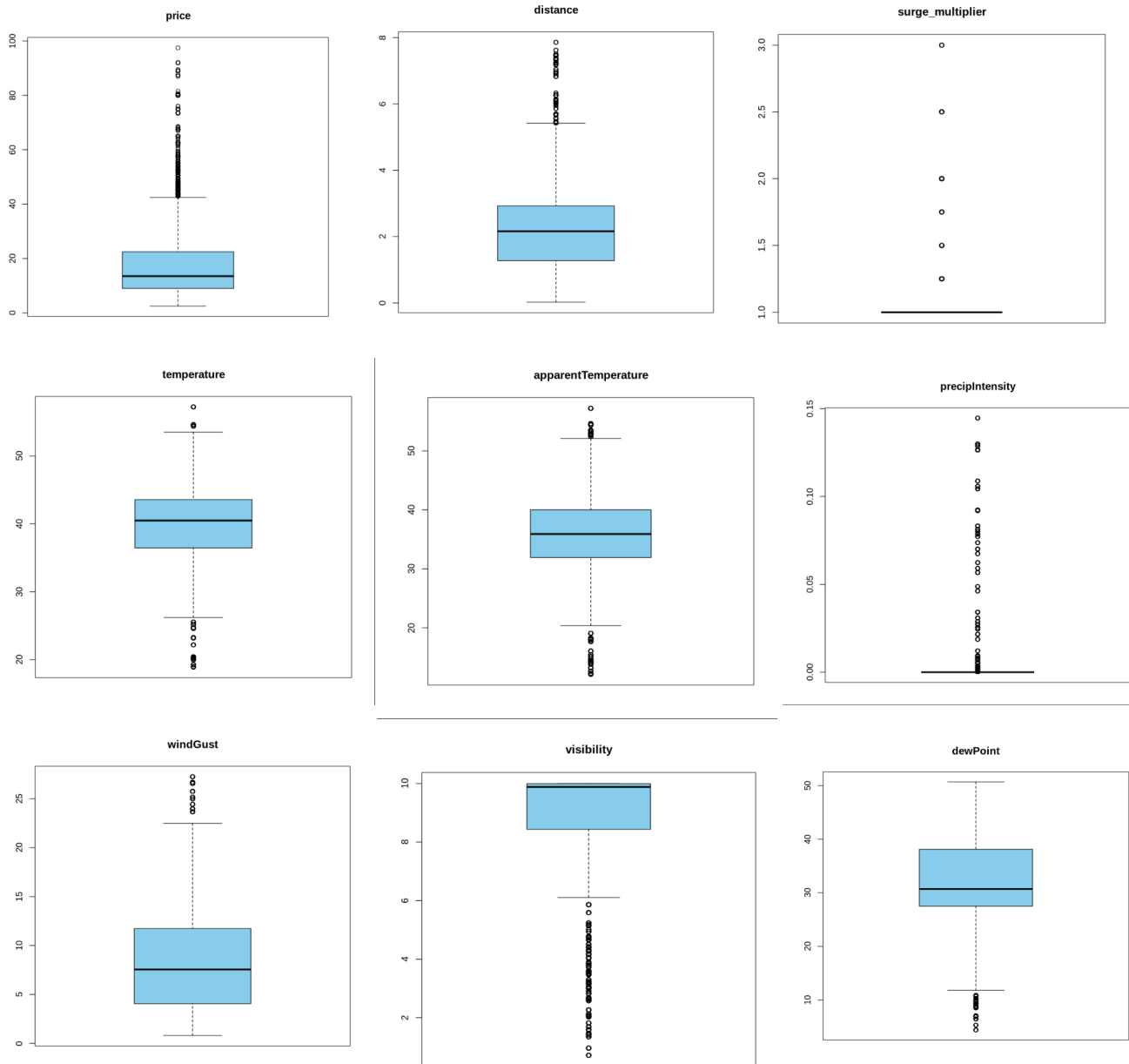
Section 2: Exploratory Data Analysis

Section 2.1: Outliers

Using IQR values the outliers were printed for each feature to get an idea of the outliers. For more information, we created a table with their number and percentage for each feature. Time related features had no outliers, while weather features had the most, with 'temperatureHigh' having the highest percentage 23.64%. Since we were interested in outlier behavior, especially extreme weather and its impact on price, we decided to keep the outliers and remove them during the implementation of models that would be affected by them. Outliers in this context represent a real phenomenon, thus are of value to our analysis.

Section 2.2: Features Type and Box Plot Visualization

We created a table to check feature type to know if Data type conversion would be necessary and the resulting data types were integer, numeric, and character, thus no conversion was necessary. Additionally, we created a table of the number of unique values with each feature to get an idea of what features would be good for classification tasks. With these two tables in mind we created a simple Box Plot Visualization of the numeric/integer features. These Box Plots helped in the visualization of outliers within each feature. To avoid errors in the graphs all the integer values were transformed into numeric. Overall, time graphs had no outliers, while price and distance showed positive skewness, variability above median and clear case of outliers. In cases like surge multipliers we see that it has discrete values with 1 as the median. Features like temperature and apparent temperature showed signs of symmetrical skewness with outliers above and below median, apparent temperature, however, had more clear outliers. Precipitation intensity showed that the median is zero and cases where there is precipitation are considered outliers. Another interesting plot was the one for visibility considering the high median value and the considerably small outlier values.



Section 2.2: Univariate Analysis

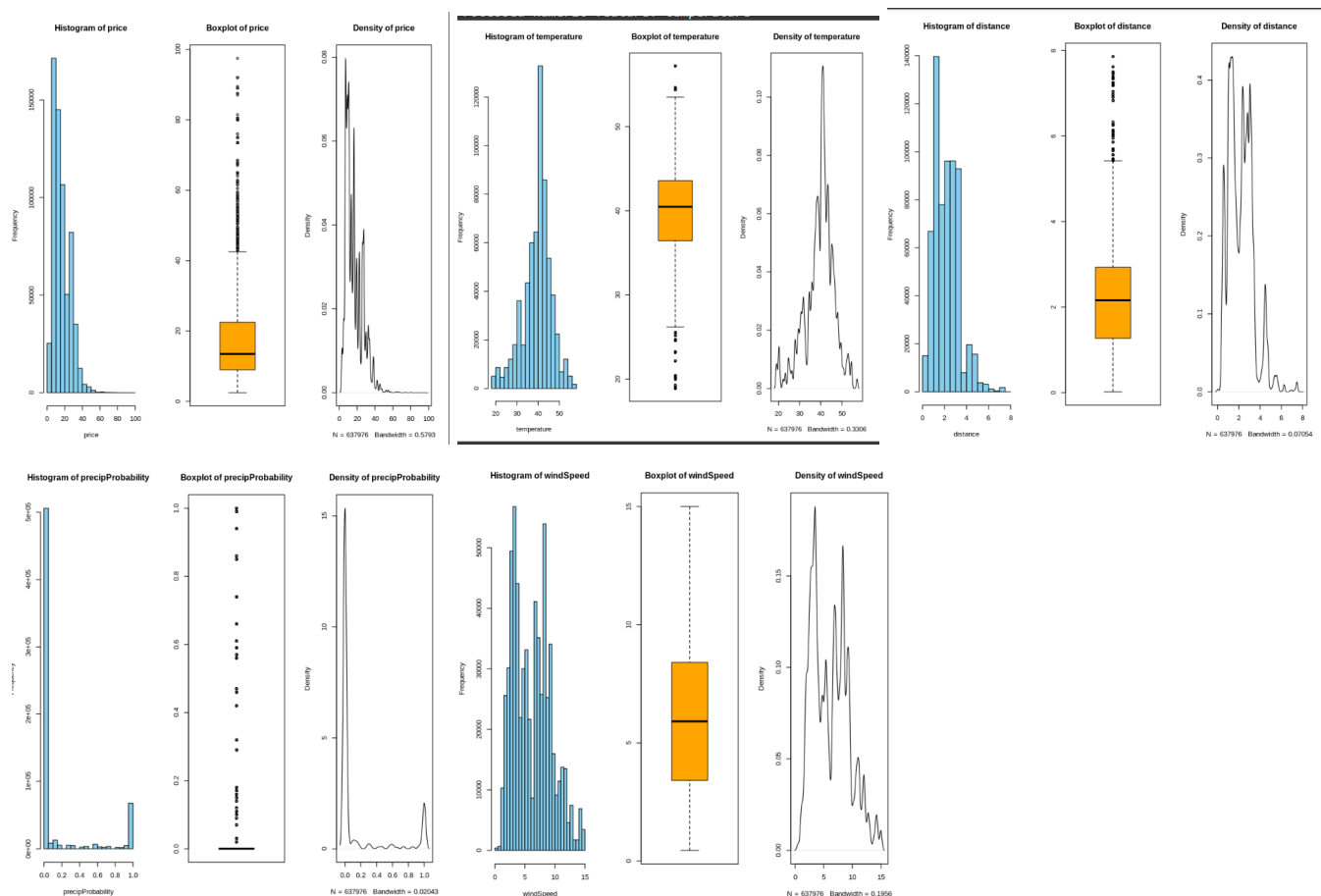
Variability for numerical columns

Another table was created to check variability of numerical features focusing on Min, Max, Range, Variance, and Standard Deviation. With values like longitude having the lowest range, while sunset and sunrise time having the highest. Another table also included CV percentage as part of the summary.

Histogram, Box Plot, and Density Plot for numerical columns

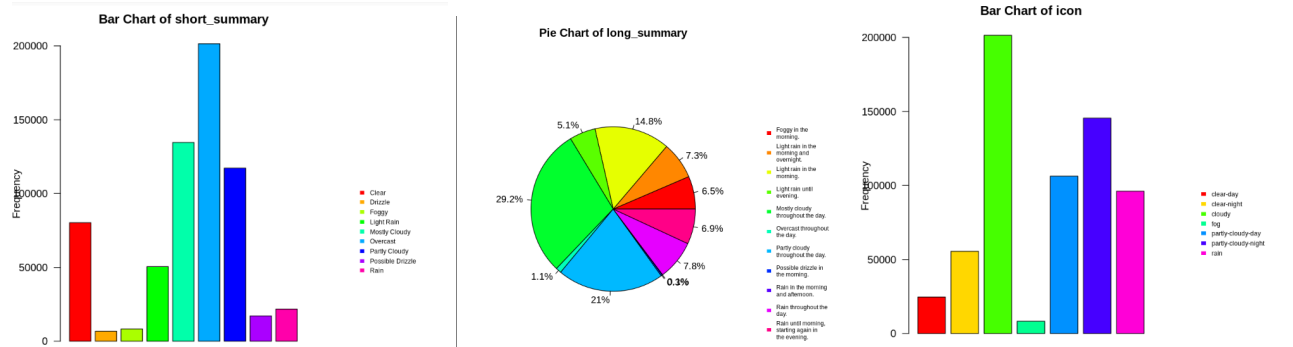
In this section a loop was created to plot a Histogram, Box Plot, and a density Plot for the numerical features. With the histogram we can observe more clearly the skewness in features like Price and

distance, while we can see how temperature is much more symmetric compared to them. Features like windSpeed have extreme values distributed over the histogram while PrecipProbability has its highest value at 0.



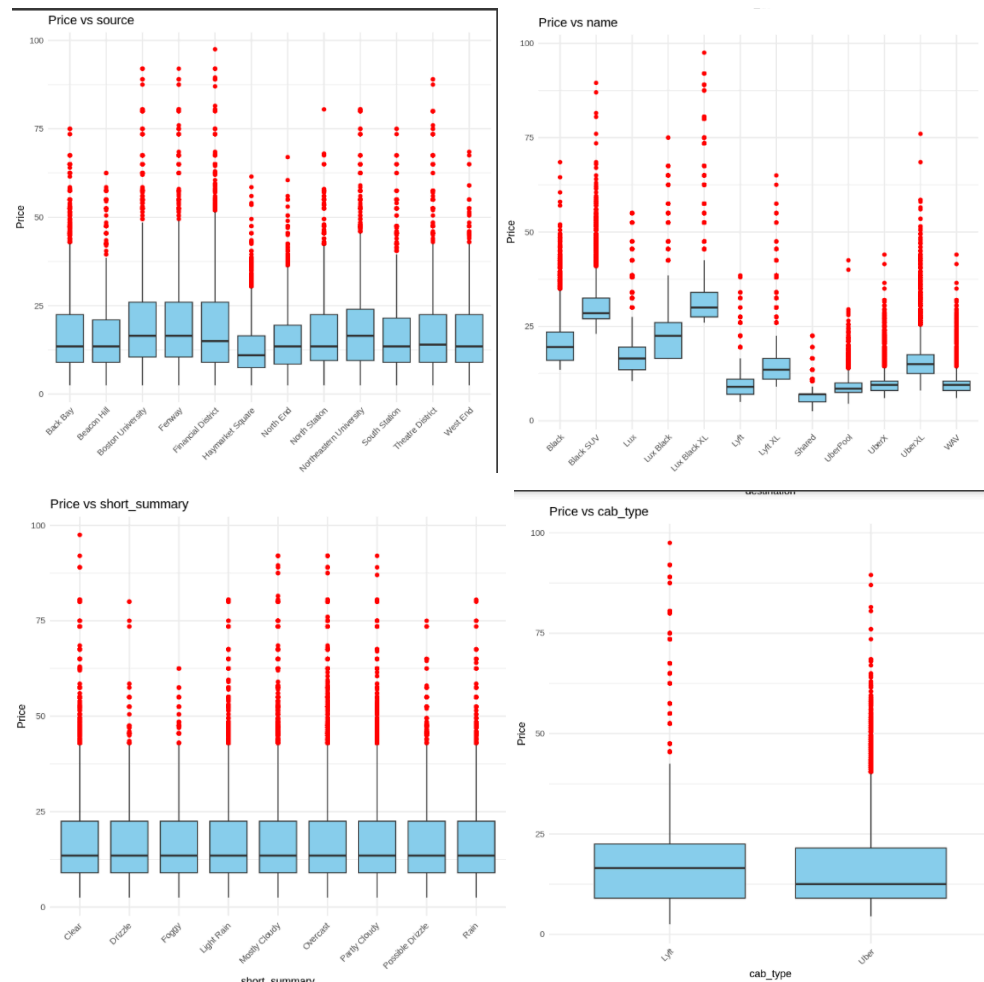
Categorical Pie Charts and Bar Charts

For all the categorical features, we included both pie and bar charts. The most interesting charts highlighted uneven distributions, providing valuable insights into the dataset. While features like Source, Destination, Name, and Cab Type showed an even distribution across attributes, others such as Short Summary, Long Summary, and Icon had less frequent values. For example, Short Summary contained fewer instances of weather conditions like rain, drizzle, fog, and others that might influence price and transportation. A similar trend was observed in Long Summary, where rain appeared only 0.3% of the time. Interestingly, Icon, which generalizes the weather, had 'cloudy' as the least frequent condition. From these graphs, we can also see that weather trends like cloudy and overcast weather are less likely to be major contributors, as they are common in the dataset's context.



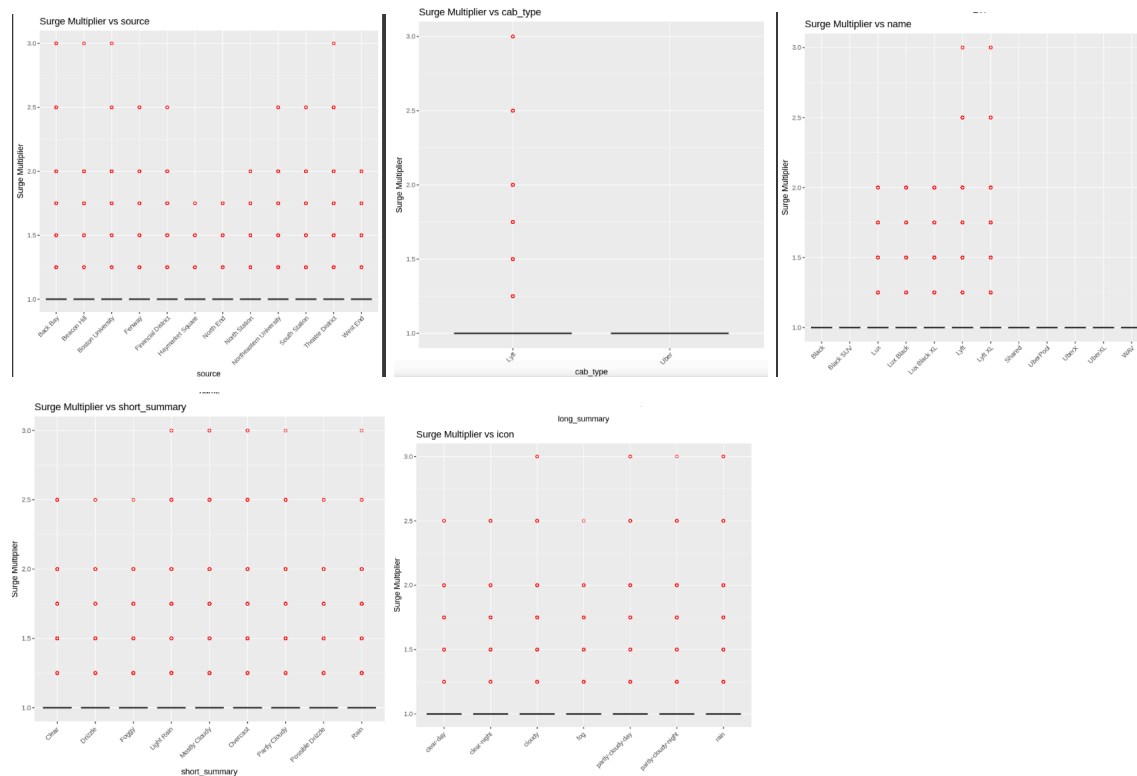
Price vs Categorical Features

Here, we decided to observe Box Plots of the Categorical Features vs Price to see how the Price differs from attribute value to another. The most interesting observations were related to outliers being associated with specific places in the Source graph. Another interesting observation was more outliers for Uber compared to Lyft. From the name feature, we observe that some types of taxi are associated with higher prices, specifically Luxury taxi options. In short summary, we can see that the taxi price within outliers in overcast and potential drizzle weather tend to be higher.



Categorical vs Surge Multiplier

In this section we do what we did with price but with surge multiplier instead. In all features the median was 1, which is normal rate, no surge multiplier. Here we were able to tell the conditions which had the highest/higher surge multiplier rate (2.5 and 3). In source, for example, some locations, like Black Bay and Boston University. From cab_type, we notice that Lyft implements surge_multiplier while Uber does not. Even some cab names are more likely to have high surge multipliers like Lyft XL. Through short summary and Icon we can see that in cases rainy, cloudy, and overcast weather the higher the surge multiplier is, namely 3 in these case.



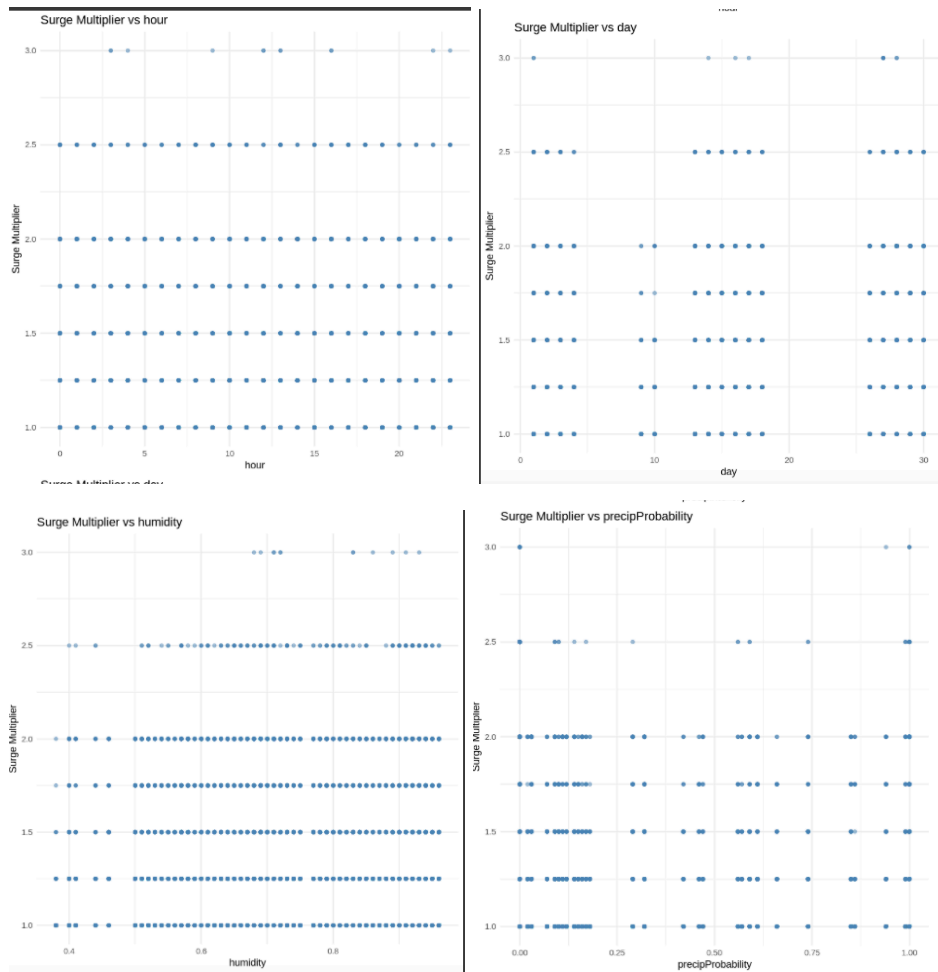
Price vs Numerical Scatter Plots

Similar to the categorical plots, we implemented numerical plots vs price. Overall, the pattern was uniform across time features like hour, day, month. However, despite the dense plot, in distance we can see an increase pattern where the price increases at high distances and for surge multipliers we see a similar pattern. Most interesting graph was the visibility because we assumed that less visibility will be associated with high prices, yet it was associated with lower prices compared to high visibility where most of the high prices were.



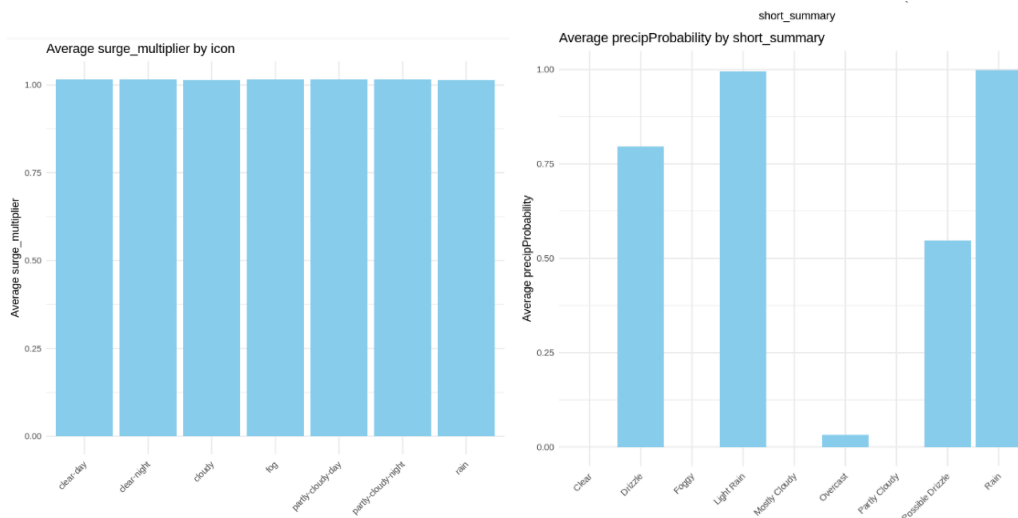
Surge Multiplier vs Numerical Features Scatter Plots

We created similar scatter plots for surge multipliers where we could see that things like certain hours had more extreme surge rate, which the common was to assign surge multipliers, during busy hours. Even certain days had the same thing. We could also see that in days where perception probability is near 1, higher surge multipliers are more common. A similar pattern is observed in high humidity.



Frequency Histograms and Weighted Graphs

Aside from frequency histograms, we created a weighted graph with a loop that does different combinations of groupings to observe different relationships. However, we did not find relevant or interesting information from them, especially compared to other graphs. Especially, since the behavior was expected in the graphs.



Classification Exploration

For this part, we examined the distribution of attribute values in categorical features for potential classification tasks. We then balanced the features with imbalanced classes.

Surge Multiplier	Category	Frequency
1	1 or Below	617001
2	Above 1	20975

Cab	Type	Frequency
1	Lyft	307408
2	Uber	330568

Surge Multiplier	Frequency
1	617001
2	11085
3	5065
4	2420
5	2239
6	154
7	12

	Name	Frequency
1	Black	55095
2	Black SUV	55096
3	Lux	51235
4	Lux Black	51235
5	Lux Black XL	51235
6	Lyft	51235
7	Lyft XL	51235
8	Shared	51233
9	UberPool	55091
10	UberX	55094
11	UberXL	55096
12	WAV	55096

Section 2.3: Statistical Tests

Kolmogorov-Smirnov Test and Anderson-Darling test

The Kolmogorov-Smirnov (KS) test and the Anderson-Darling (AD) test both assess the normality of data but in different ways. The KS test compares the empirical cumulative distribution of the data to a normal distribution, while the AD test gives more weight to the tails of the distribution, making it more sensitive to deviations in the tails. In our results, both tests returned similar p-values, mostly 0, indicating a clear deviation from normality for all features. However, the AD test, due to its sensitivity to tails, might have identified non-normality slightly more robustly in some cases, although the overall results were consistent between the two tests. This section also contained an analysis of the skewness and Kurtosis.

```
Normally Distributed Features:
NULL

Not Normally Distributed Features:
[1] "hour"           "day"
[3] "month"          "price"
[5] "distance"       "surge_multiplier"
[7] "latitude"       "longitude"
[9] "temperature"    "apparentTemperature"
[11] "precipIntensity" "precipProbability"
[13] "humidity"       "windSpeed"
[15] "windGust"       "visibility"
[17] "temperatureHigh" "temperatureLow"
[19] "apparentTemperatureHigh" "apparentTemperatureLow"
[21] "dewPoint"       "pressure"
[23] "windBearing"    "cloudCover"
[25] "uvIndex"        "ozone"
[27] "sunriseTime"    "sunsetTime"
[29] "moonPhase"      "precipIntensityMax"
[31] "temperatureMin" "temperatureMax"
[33] "apparentTemperatureMin" "apparentTemperatureMax"
```

Wilcoxon Signed-Rank Test & Spearman Correlation Test

We couldn't perform the paired t-test because the data did not meet the assumption of normality. Also, the Shapiro-Wilk test was not suitable due to the large sample size, which can lead to misleading results. Instead, we used the Wilcoxon Signed-Rank Test, which is a non-parametric test that does not assume normality, making it more suitable for our data. While the paired t-test requires numeric values for both variables, the Wilcoxon test handles situations where normality is violated, as apparent from previous tests. The Wilcoxon Signed-Rank Test was used instead of the Wilcoxon rank-sum test because the latter assumes independence, which we assumed was not applicable in our paired data. We choose specifically features like temperature and apparent temperature to see if there was significance between the two and if one of them would be enough as a feature, and even predictor.

Across all comparisons, the Wilcoxon Signed-Rank Test shows that the medians of actual temperatures and their corresponding apparent temperatures differ significantly, suggesting that people perceive temperature differently than measured values in all cases.

Since the result was different to our expectation we wanted to do another test to explore significant relationships. The most interesting pairs were the ones comparing actual vs apparent temperature, wind metrics, and precipitation. These pairs showed strong and statistically significant correlations (rho values close to 1, p-value = 0). As a result, this indicates a consistent and predictable relationship between

measured and perceived weather features. In contrast, price had no significant correlation with geographic coordinates like latitude and longitude.

The Wilcoxon Signed-Rank Test showed significant differences in medians between actual and apparent temperatures, while the Spearman Correlation revealed strong monotonic relationships. Thus, the variables differ in value but move in the same direction and follow the same trend. Hence, there is a potential for one of the pair values to be enough when used in data science models.

Chi-squared Test

In this section, we performed Chi-Square tests on all unique pairs of categorical variables in the `cleaned_data` dataset to check for statistical independence. A contingency table was built for each pair, then we categorized the results into significant (dependent) and independent pairs, and printed them for easy interpretation. Additionally, we printed a table with relevant values and made a conclusion about their relationship.

		Variable1	Variable2	Chi_Square	p_value
--- Significant (Dependent) Variable Pairs ---					
1 .	cab_type vs long_summary	source	destination	639065.10	0.0000
2 .	cab_type vs short_summary	source	short_summary	261.07	0.0000
3 .	destination vs icon	source	long_summary	292.05	0.0000
4 .	destination vs long_summary	source	icon	220.23	0.0000
5 .	destination vs short_summary	destination	short_summary	299.97	0.0000
6 .	icon vs source	destination	long_summary	327.55	0.0000
7 .	long_summary vs source	destination	icon	288.42	0.0000
8 .	short_summary vs source	cab_type	name	637976.00	0.0000
		cab_type	long_summary	37.33	0.0000
		short_summary	long_summary	541196.69	0.0000
		short_summary	icon	2552093.15	0.0000
		long_summary	icon	398233.51	0.0000
		cab_type	short_summary	24.43	0.0019
		cab_type	icon	10.82	0.0942
		name	long_summary	87.46	0.9443
		name	short_summary	65.21	0.9673
		name	icon	39.76	0.9957
		source	cab_type	2.10	0.9981
		destination	cab_type	1.99	0.9985
		source	name	2.10	1.0000
		destination	name	1.99	1.0000
--- Independent Variable Pairs ---					
1 .	cab_type vs destination				
2 .	cab_type vs icon				
3 .	cab_type vs source				
4 .	destination vs name				
5 .	icon vs name				
6 .	long_summary vs name				
7 .	name vs short_summary				
8 .	name vs source				

Fisher's Exact Test was additionally used to determine if there is a non-random association between two categorical variables, especially when sample sizes are small or expected frequencies are low. The Fisher's method results across all variable pairs indicate statistically significant differences in both means and variances ($p\text{-value} = 0$ in all paired t-tests and F-tests). This suggests that although the compared features may seem related (e.g., temperature vs. apparent temperature), they behave differently both in central tendency and in how spread out their values are. Therefore, each variable provides distinct information and should not be treated as redundant in modeling or analysis. We created code to determine the necessity of the test and it wasn't that necessary in most cases. We also did a table to observe if the conclusion changes with and without the test.

Comparison: temperatureMax vs apparentTemperatureMax

Paired t-test (Difference in Means):

t = 1156.052 | p-value = 0

F-test (Ratio of Variances):

F = 0.6623 | p-value = 0

--- Interpretation ---

Means: Significant difference → Variables behave differently.

Variances: Significant difference → One may be more variable; assess based on domain or model needs.

Variable1	Variable2	Chi_Square	p_value	Adjusted_p_value	Conclusion	Fisher_statistic	Conclusion_Change
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<chr>
source	destination	639065.10	0.0000	0.0000	Dependent (Reject H0)	0	No Change
source	cab_type	2.10	0.9981	0.9981	Independent (Fail to Reject H0)	0	No Change
source	name	2.10	1.0000	1.0000	Independent (Fail to Reject H0)	0	No Change
source	short_summary	261.07	0.0000	0.0000	Dependent (Reject H0)	0	No Change
source	long_summary	292.05	0.0000	0.0000	Dependent (Reject H0)	0	No Change

Variable1	Variable2	Chi_Square	p_value	Conclusion	Fisher_statistic
<chr>	<chr>	<dbl>	<dbl>	<chr>	<dbl>
1 source	destination	639065.	0	Dependent (R...	NA
2 source	cab_type	2.1	0.998	Independent ...	NA
3 source	name	2.1	1	Independent ...	NA
4 source	short_summary	261.	0	Dependent (R...	NA
5 source	long_summary	292.	0	Dependent (R...	NA
6 source	icon	220.	0	Dependent (R...	NA
7 destination	cab_type	1.99	0.998	Independent ...	NA
8 destination	name	1.99	1	Independent ...	NA
9 destination	short_summary	300.	0	Dependent (R...	NA
10 destination	long_summary	328.	0	Dependent (R...	NA
11 destination	icon	288.	0	Dependent (R...	NA
12 cab_type	name	637976	0	Dependent (R...	NA
13 cab_type	short_summary	24.4	0.0019	Dependent (R...	NA
14 cab_type	long_summary	37.3	0	Dependent (R...	NA
15 cab_type	icon	10.8	0.0942	Independent ...	NA
16 name	short_summary	65.2	0.967	Independent ...	NA
17 name	long_summary	87.5	0.944	Independent ...	NA
18 name	icon	39.8	0.996	Independent ...	NA
19 short_summary	icon	2552093.	0	Dependent (R...	NA
20 long_summary	icon	398234.	0	Dependent (R...	NA

Section 2.3: Bivariate and Multivariate Analysis

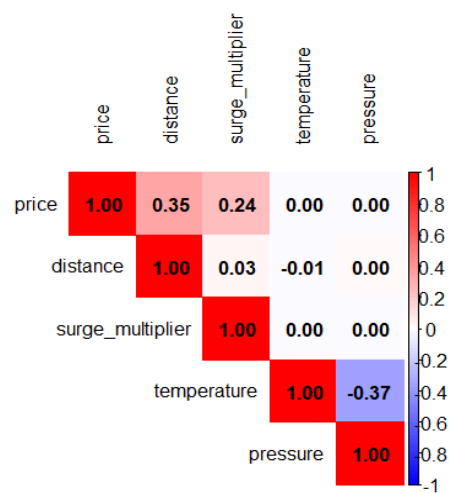
Multivariate and Bivariate Analysis are essential subsections of Exploratory Data Analysis (EDA), focusing on uncovering deeper insights by exploring relationships between variables. In this section, both techniques will be explained in a way to identify patterns, dependencies, and interactions within the dataset.

Starting with bivariate analysis, we examine two variables simultaneously using correlation matrices, scatter plots, and cross tabulations, allowing us to investigate connections between key variables like distance, price, and surge_multiplier. This helps detect potential linear or non-linear relationships that can inform further analysis.

Moving into multivariate analysis, we extend our scope to multiple variables simultaneously. We utilize comprehensive heatmaps to visualize correlations across broader numerical datasets and time series analysis to track fluctuations in key metrics like temperature and surge_multiplier over time.

Additionally, we will explore probability distributions using binomial and multinomial models to understand the frequency and spread of specific outcomes, and we compile summary tables to break down data by categories such as cab_type and hour. Further statistical analysis includes the coefficient of variation to assess data volatility, covariance analysis to determine directional relationships, and decision tree modeling to identify key predictors impacting surge_multiplier. By systematically analyzing these variables within the EDA framework, we aim to build a well-rounded understanding of how different data points connect, interact, and influence one another, setting the stage for more targeted predictive analysis and data-driven decision-making.

Correlation Matrix: The correlation matrix shows how the main variables: price, distance, surge_multiplier, temperature, and pressure, are connected.



The correlation matrix provides a comprehensive overview of the relationships between key variables, offering valuable insights into how they interact with one another. The strongest positive correlation identified is between price and distance (0.35), indicating that as the distance of a ride increases, the price tends to rise as well. While this relationship is moderately positive, it is not particularly strong, suggesting that other factors beyond distance play a significant role in determining ride prices. Similarly, the surge multiplier shows a weak positive correlation of 0.24 with price, indicating that surge pricing slightly elevates the cost of rides, but the impact is limited. Interestingly, the relationship between temperature and pressure stands out with a moderate negative correlation of -0.37. This aligns with typical meteorological patterns where rising temperatures are often associated with decreasing atmospheric pressure. This is the strongest correlation observed in the dataset, highlighting a more defined interaction between weather-related variables. However, most of the other correlations are close to zero, suggesting little to no linear relationship between those variable pairs. For instance, price and temperature as well as price and pressure both show near-zero correlations, indicating that fluctuations in weather conditions do not directly impact ride pricing in the current dataset. Similarly, the correlation between distance and surge multiplier is nearly negligible (0.03), implying that distance alone is not a significant driver of surge pricing. While some moderate correlations exist, particularly between price and distance, and temperature and pressure, the majority of the relationships are weak. This suggests that the current set of features may not be particularly strong predictors of ride prices. To improve model performance, it might be beneficial to consider additional variables, such as time of day, ride duration, traffic conditions, or contextual factors like major events or holidays that could potentially influence both pricing and surge multipliers.

Covariance matrix:

	hour	day	month	price	distance
hour	47.76	5.10	-0.30	-0.17	0.03
day	5.10	98.88	-4.12	-0.26	0.06
month	-0.30	-4.12	0.24	0.01	0.00
price	-0.17	-0.26	0.01	87.14	3.63
distance	0.03	0.06	0.00	3.63	1.28
surge_multiplier	0.00	0.00	0.00	0.23	0.00
latitude	0.01	0.01	0.00	0.00	0.00
longitude	0.00	0.01	0.00	0.00	0.00
temperature	10.81	-5.41	-0.32	0.29	0.00
apparentTemperature	11.51	-16.87	0.00	0.28	-0.02
precipIntensity	-0.04	0.03	0.00	0.00	0.00
precipProbability	-0.27	0.15	-0.02	0.00	0.00
humidity	-0.26	-0.10	0.00	0.01	0.00
windSpeed	1.71	13.43	-0.48	0.09	0.02
windGust	2.75	20.67	-0.65	0.16	0.05
visibility	3.14	2.18	0.05	-0.03	0.01
temperatureHigh	0.37	-16.42	0.13	0.26	-0.02
temperatureLow	1.16	-7.55	-0.36	0.12	0.00
apparentTemperatureHigh	0.62	-29.12	0.44	0.33	-0.04
apparentTemperatureLow	1.47	-16.85	-0.32	0.16	-0.02
dewPoint	1.06	-8.32	-0.44	0.49	-0.03
pressure	-5.88	-52.44	3.53	-0.39	-0.04
windBearing	17.95	58.65	-4.86	0.79	-0.25
cloudCover	0.06	-0.18	0.00	0.01	0.00
uvIndex	1.07	-0.08	0.00	-0.02	0.00
ozone	4.31	89.42	-1.02	0.27	0.32
sunriseTime	-165287.06	-2109669.49	254708.24	-2285.75	4519.69

The covariance matrix gives us a snapshot of how different numerical variables in our dataset move together. It's like a cheat sheet that tells us whether two variables tend to increase or decrease together or if they don't have much of a relationship at all. Starting with price and distance, we see a positive covariance, meaning that as the distance of a ride goes up, the price generally goes up too. That makes sense, longer rides are naturally more expensive. However, when we look at price and temperature, the covariance is slightly negative. This suggests that when temperatures rise, prices tend to dip a bit. This could indicate that fewer people are requesting rides when it's hotter outside, or maybe people are opting for alternative transportation options. For distance and surge_multiplier, there's a moderate positive covariance, indicating that longer rides are more likely to have a surge multiplier applied. This could be because longer rides tie up drivers for more time, increasing demand for other rides and triggering surge pricing. Interestingly, the relationship between temperature and surge_multiplier is negative, suggesting that hotter weather might reduce surge pricing. Maybe people are staying indoors more, leading to less demand for rides. The covariance between windSpeed and surge_multiplier is slightly positive, hinting that surge pricing might tick up a bit when it's windier. This could be due to more challenging driving conditions or people opting for rides instead of walking. Similarly, humidity and surge_multiplier show a positive covariance, suggesting that on more humid days, surge pricing is slightly more common. This might indicate that people are less willing to walk or wait for public transport when the weather is muggy. Lastly, the covariance between pressure and surge_multiplier is slightly negative, meaning that higher pressure (often associated with more stable, clear weather) might coincide with lower surge prices.

Variation Coefficient:

	cab_type	mean_price	sd_price	cv
	<chr>	<dbl>	<dbl>	<dbl>
1	Lyft	17.4	10.0	0.577
2	Uber	15.8	8.52	0.541

The coefficient of variation (CV) analysis provides valuable insights into the variability of prices for both Lyft and Uber. Lyft, with a mean price of 17.4 and a standard deviation of 10.0, has a CV of 0.577, indicating that its prices are relatively more variable, with a 57.7% spread relative to the mean. This suggests that Lyft experiences more pronounced price fluctuations, potentially due to factors such as surge pricing, demand peaks, or route variability. On the other hand, Uber, with a mean price of 15.8 and a standard deviation of 8.52, exhibits a slightly lower CV of 0.541, reflecting a 54.1% variability relative to the mean. This indicates that Uber's pricing is somewhat more stable and less susceptible to extreme price swings compared to Lyft. Overall, the analysis reveals that while both services demonstrate significant price variability, Lyft's pricing is slightly more unpredictable, which could have implications for forecasting surge pricing and modeling price sensitivity.

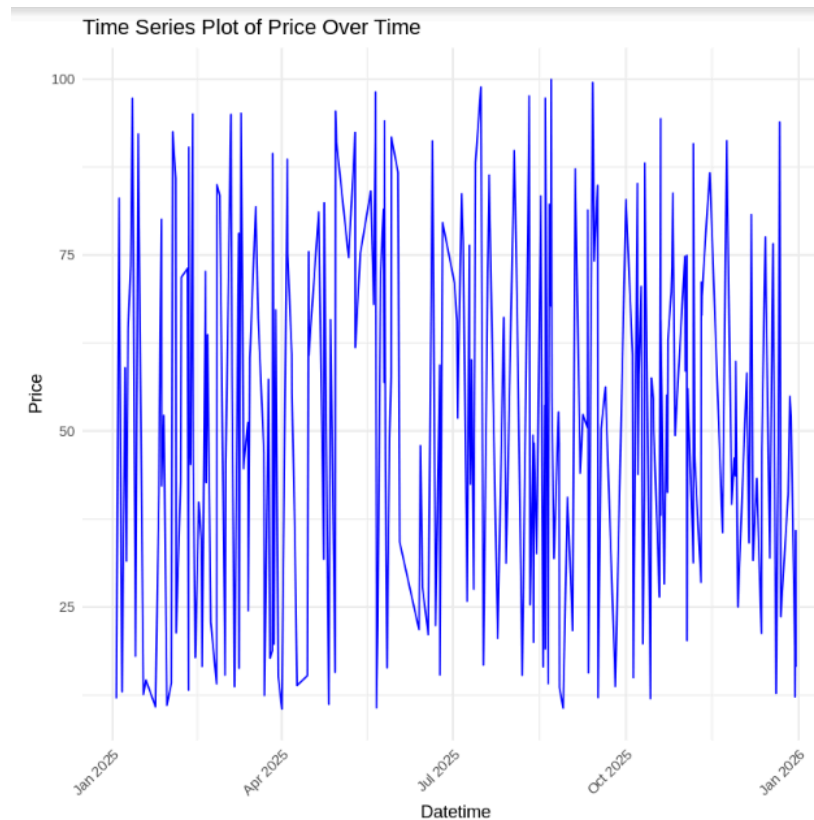
Frequency Table:

	cab_type	surge_multiplier	n	probability
1	Lyft	1.00	27282	0.4537018559
2	Lyft	1.25	1105	0.0183762389
3	Lyft	1.50	576	0.0095789264
4	Lyft	1.75	229	0.0038082884
5	Lyft	2.00	224	0.0037251380
6	Lyft	2.50	10	0.0001663008
7	Uber	1.00	30706	0.5106432515

The frequency table output provides a breakdown of the surge multiplier occurrences for Lyft and Uber rides, along with their respective probabilities. For Lyft, the most common surge multiplier is 1.0, accounting for 27,282 rides, which represents approximately 45.37% of all Lyft rides. This indicates that nearly half of the Lyft rides occur without any surge pricing. However, surge multipliers increase in frequency as they rise, but at much lower rates. For instance, the 1.25 surge multiplier occurs in 1,105 rides, representing 1.88% of Lyft rides, while the highest surge multiplier of 2.5 is rare, appearing in only 10 rides, making up a mere 0.016% of all Lyft rides. In contrast, Uber has a single entry for a surge multiplier of 1.0, occurring in 30,706 rides, which constitutes approximately 51.06% of all Uber rides. This suggests that Uber rides experience less surge pricing variability compared to Lyft, with a majority of rides occurring without any surge pricing. Overall, the distribution shows that while Lyft exhibits more surge pricing variability with multiple surge multiplier levels, Uber remains more consistent, predominantly operating at a surge multiplier of 1.0. This difference could indicate distinct pricing

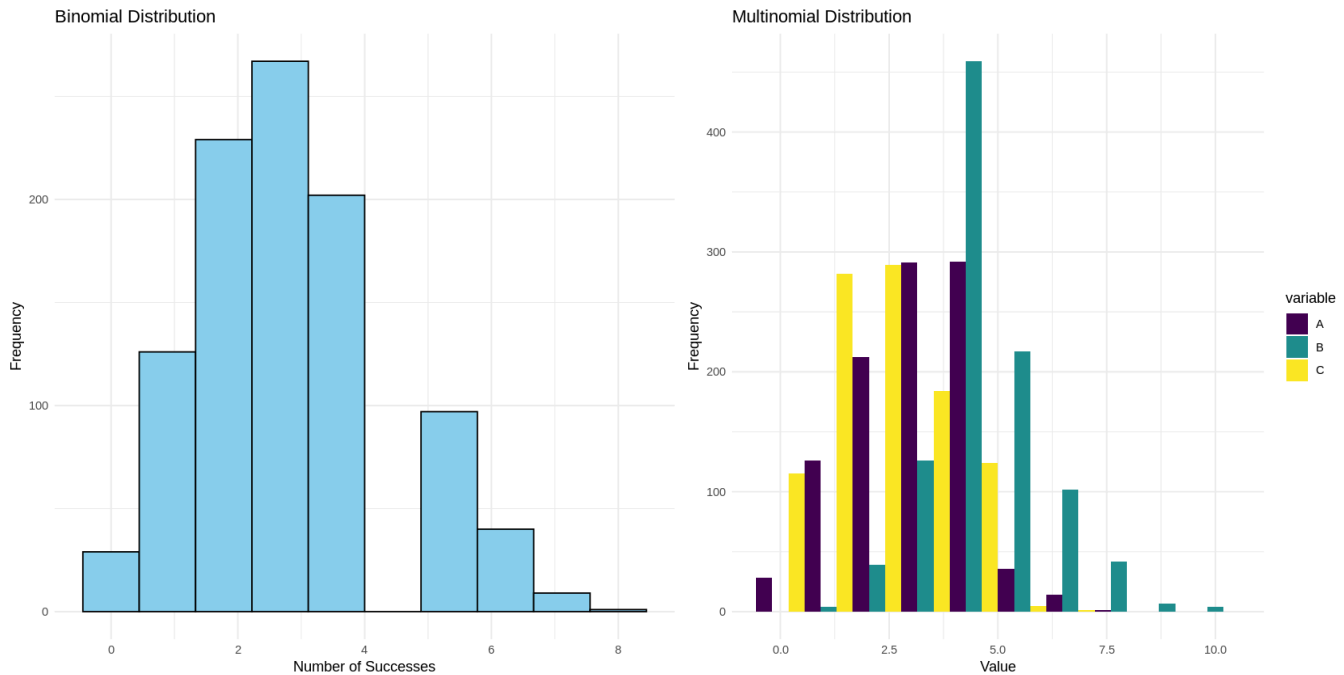
strategies, with Lyft potentially leveraging surge pricing more aggressively in response to demand spikes.

Time Series plot:



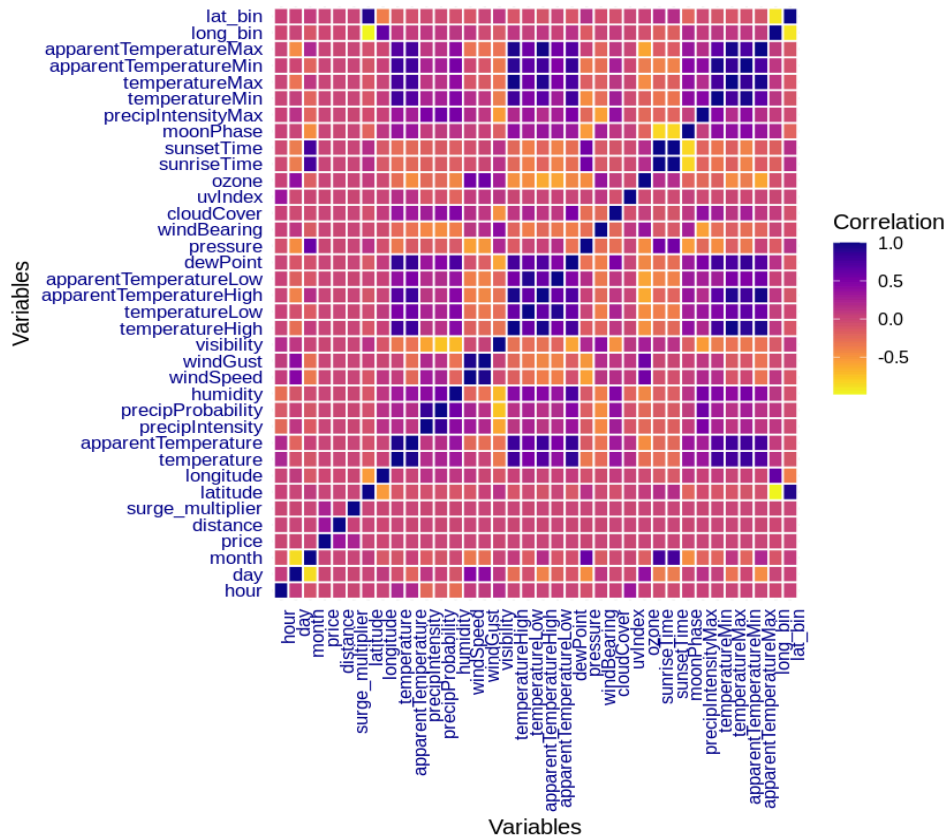
The time series plot illustrates the variation in price over a one-year period from January 2025 to January 2026, with prices ranging from approximately 10 to 100 units. The data exhibits high volatility with frequent and substantial fluctuations throughout the entire timeline, indicating a lack of a discernible long-term upward or downward trend. Recurring spikes and dips are noticeable at the start of each quarter, suggesting potential seasonal or cyclical patterns that warrant further investigation. However, the erratic nature of the data makes it challenging to identify specific seasonal trends without additional smoothing or decomposition techniques. The consistent amplitude of fluctuations highlights a dynamic dataset potentially influenced by external factors, such as demand changes, weather conditions, or surge pricing. Additionally, several sharp peaks and deep troughs stand out as potential outliers, which may require further scrutiny to assess their origins and impact on overall trends. To gain deeper insights, applying moving averages, decomposing the time series into trend and seasonal components, and implementing outlier detection methods could provide a more comprehensive understanding of the observed price variations.

Probability Distributions:



The binomial distribution shown in the first graph is characterized by its typical bell-shaped curve, indicating the frequency of "successes" across several trials. Here, we observe that most occurrences cluster around 2 to 4 successes, with a noticeable drop-off as the number of successes increases or decreases. This distribution is asymmetrical, showing a slightly right-skewed pattern, suggesting that most observations are concentrated around lower success counts. This aligns with typical ride-share scenarios where extreme surge multipliers are less frequent, and most rides occur under more common conditions. On the other hand, the multinomial distribution depicted in the second graph provides a broader view by considering multiple categories or outcomes simultaneously. The three distinct bars represent different categories (A, B, C), where category A represents a specific surge multiplier range or a specific time of day (e.g., morning, afternoon, evening), where the frequency of rides within that segment is being counted and category B indicates a particular cab type (e.g., Lyft or Uber) or a specific weather condition that influences ride frequency, such as "Clear" vs. "Rainy", while category C represents another significant factor, such as ride distance range (e.g., short, medium, long trips) or a specific source/destination classification. As shown in the second figure, Category B has a more pronounced peak around the value of 5, indicating that it is the most common outcome in that range. Categories A and C display a more dispersed pattern, indicating a broader spread of occurrences. This kind of distribution is particularly useful in ride-share data analysis, where different surge multipliers or cab types can be categorized and analyzed collectively, offering a comprehensive view of how various factors interact within the dataset.

Heat Map: The heatmap provides a visual snapshot of how various numerical variables in the dataset are connected, as shown in the figure below.



The standout relationship is the strong positive correlation between temperature and apparent temperature, indicating that as the temperature rises, the perceived temperature also increases, which makes sense given their inherent connection. Another noticeable pattern is the positive correlation between wind speed and wind gust, which suggests that higher wind speeds often coincide with stronger gusts. There's also a moderate positive link between temperature and humidity, highlighting that warmer temperatures tend to be associated with more humid conditions. This factor could influence ride demand and pricing. On the other hand, the slight inverse correlation between distance and surge multiplier hints that surge pricing might be more prevalent for shorter rides, potentially due to higher demand in densely populated areas. Additionally, pressure and temperature show a mild negative correlation, suggesting that higher pressure systems could be linked to cooler conditions. These insights not only highlight the interdependencies among variables but also point to potential factors that could play a significant role in predictive modeling, such as surge pricing, weather conditions, and ride distances.

Two-way ANOVA:

(In this section the original sampling of $X = 250$ was used.)

To explore the influence of various factors on the variables, we conducted multiple two-way ANOVA table tests. Each test assessed the main effects and interaction effects of two categorical independent variables on a continuous variable. In this section we did the anova tests based on what we thought could have a useful and interesting interaction between the factors and the variables. Additionally while doing the tests we found some tables that had no interaction and put them in a discarded pile as shown in the code. The tables that gave us some interesting results had the following:

Continuous variable	Factor A	Factor B	Results
Price	source	destination	Both factors and their interaction significantly affected price ($p < 2e-16$), suggesting that both where a ride starts and ends strongly influence cost, with varying effects across combinations.
distance	source	destination	Both factors and their interaction were highly significant ($p < 2e-16$), indicating distance varies widely by pickup and drop-off combinations.
	source	cabtype	Strong source effect and significant interaction ($p < 2e-16$), but <i>cab_type</i> alone was not significant ($p = 0.433$), therefore cab types may serve areas differently.
	Hour	Destination	Significant destination ($p < 2e-16$) and interaction ($p = 0.020$), though hour alone wasn't significant ($p = 0.353$) suggesting destination matters more, but the effect of time varies slightly across destinations.
Temperature & Apparent temperature	source	hour	For both tests of temperature and apparent temperature, only <i>hour</i> ($p < 2e-16$) and the interaction were significant ($p(T) < 8.22e-05$, $p(AT) = 7.18e-05$), but not source ($p(T) = 0.179$, $p(AT) = 0.164$), meaning temperature varies strongly over the day, with small variations across sources.
Temperature	Short summary	hour	all effects significant ($p < 2e-16$). This implies weather summaries (e.g., "Rain", "Clear") and time of day are strong predictors of temperature.
visibility	source	Long summary	<i>Long_summary</i> was highly significant ($p = < 2e-16$), so was the interaction ($p = 0.0274$), the source was also significant ($p = 0.0374$) suggesting visibility is most impacted by specific weather types, with variation across sources.
Wind speed	month	Short summary	All effects were highly significant ($< 2e-16$), indicating that wind patterns vary by weather condition and month,

			with notable interaction effects.
ozone	month	Short summary	Strong effects from both factors and their interaction ($p < 2e-16$) suggest that ozone levels vary greatly with both time of year and weather conditions.

From our discarded pile an unexpected result we found was that “hour” and “day” have no effect on “price” you would usually expect time-related factors to significantly affect price due to rush hours, weekend and weekdays. However the anova table indicated otherwise. It showed that mainly source, destination, and cab type are the main factors that affect price instead.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
source	11	186931	16994	221.0	<2e-16	***
destination	10	156994	15699	204.2	<2e-16	***
source:destination	50	479185	9584	124.7	<2e-16	***
Residuals	82525	6344721	77			

Signif. codes:	0	'***'	0.001	'**'	0.01	'*' 0.05 '.' 0.1 ' ' 1

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
hour	23	1556	67.65	0.780	0.761	
day	16	695	43.46	0.501	0.948	
hour:day	292	26972	92.37	1.064	0.215	
Residuals	82265	7138607	86.78			

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
month	1	0	0	0.001	0.977	
cab_type	1	45957	45957	532.976	<2e-16	***
month:cab_type	1	107	107	1.241	0.265	
Residuals	82593	7121766	86			

Signif. codes:	0	'***'	0.001	'**'	0.01	'*' 0.05 '.' 0.1 ' ' 1

Section 3: Model Building

Section 3.1: Generalized Multiple Linear Regression Model Building

3.1.1 Introduction

This section presents a comprehensive overview of the multiple linear regression analysis conducted on the response variable “price.” It will introduce the goal of the multivariate regression analysis, the detailed experimental design, the trial setup, subsequent procedures, problems, and an in-depth analysis of the results, outcomes, and decisions concerning the development and implementation of the regression models. The main objective is to develop a model that can perform two main outcomes: prediction (i.e. R-sq predicted), and statistical inference (i.e. confidence and prediction interval). The model-building will include an in-depth analysis of the procedures used to fit the model, as well as the transformations, and five-assumption criteria analysis.

3.1.2 Sampling Set

A sample of 42 predictors from the 57 total predictors, (again, removed in the pre-analysis manual feature selection) was selected. The exact number of records is represented by an X of 32, where X represents the number of records per hour of every day of every month available in the dataset. As previously stated, this was accomplished to ensure that the sampling was both stratified and representative; particular days for the most part, each hour-day-month combination had an equal number of records, ensuring that a particular hour, day, or month does not have a significant influence over the predictive power of the model. Therefore, an X of 32 translates into exactly 10642 records, with the same 42 features. The selected X was mostly arbitrary, but selected to ensure that the sample would contain roughly 10,000 records, to not overwhelm the sampling, and to keep it at roughly the requirement of the project. Later brief sections will cover the effect of small vs. large sampling parameters for X .

3.1.3 Training, and Testing Data

We perform a 70/30 train-test split of the sampled data, meaning out of 10642 records, we will have 70% of the records for training and 30% for testing. Outlier removal could potentially be done at this point, however, we want to see the effect of the model without any processing, and the distribution of the residuals before making such decisions, as the outliers may or may not be significant enough to remove. Therefore, outlier removal will be kept for the models that receive transformations.

3.1.4 Model-Building & Analysis

This section will highlight the main overview of the model brainstorming and selection process. The most basic form of OLS methodology was applied to our cleaned dataset without any feature selection or transformations; any subsequent models or procedures used aim to significantly reduce violated assumptions or decrease the risk or degree of overfitting present in the linear model. Moreover, feature selection will be performed to reduce model complexity and overfitting.

3.1.5 Simple Linear Model: Model Summary, ANOVA, Assumptions, and Hypothesis Tests

The categorical variables will be encoded using dummy variables (automatically), and each factor of the categorical variable will represent a parameter of their own. This means that feature selection will be important, as we have features and factor levels. Moreover, we observe that the overwhelming majority of the predictors are insignificant at a 5% significance level (**TABLE 3.1**), again signifying feature selection's importance. In the ANOVA and Summary Table (Available in R Output), we can see that the significant predictors are: *name*, *source*, *destination*, *days*, *distance*, *cab_type*, and *surge_multiplier*. From a real-world perspective, these make sense, as they are directly related to price. Running an initial regression model with no feature selection (i.e. all 42 features that we included after the pre-analysis manual feature selection), outlier removal, or transformations, we can note the following observations:

- **Coefficient of Determination:** R^2 and R^2 -adjusted are calculated automatically on the summary table and yield 92.88% and 92.84% respectively. While significantly high (signifying that the model accounts for 92.88% of the variation), this is very misleading at this point, because we cannot tell if the model is overfitted or not until we calculate R^2 -predicted. Calculating the R^2 -predicted (calculated using $1 - [PRESS/TSS]$) yields 92.52%, which signifies a high predictive power, and no signs of overfitting. Yet, the model remains overly complex, and we cannot say until we see the assumptions and perform feature selection whether the model is suitable or not.
- **Multicollinearity:** The in-built `VIF()` function produced an error, hence, we created our own VIF function from the function in the multiple linear regression handout, $VIF = 1/(1-R^2_i)$ for all predictors i from 1 to 42. We see that there is a significant number of multicollinearity occurring; some show large numbers, and others show "NA" signifying that some predictors have perfect multicollinearity (Available in R Output). This will be resolved in the transformation section using Post-Analysis Feature Selection. We can get a quick idea about multicollinearity through the correlation matrix heatmap performed in the EDA as well.

TABLE 3.1: ANOVA TABLE for Basic Linear Regression Model

<u>Analysis of Variance Table</u>							
Response: price							
	Df	Sum Sq	Mean Sq	F value	Pr(>F)		
hour	1	134	134	20.8497	5.049e-06	***	
day	1	48	48	7.5197	0.006118	**	
month	1	17	17	2.5771	0.108463		
source	11	16622	1511	234.6345	< 2.2e-16	***	
destination	10	14883	1488	231.0906	< 2.2e-16	***	
cab_type	1	2320	2320	360.1878	< 2.2e-16	***	
name	10	489104	48910	7594.5290	< 2.2e-16	***	
distance	1	49390	49390	7668.9895	< 2.2e-16	***	
surge_multiplier	1	20453	20453	3175.8963	< 2.2e-16	***	
latitude	1	0	0	0.0371	0.847216		
longitude	1	1	1	0.1254	0.723310		
temperature	1	1	1	0.0990	0.753077		
apparentTemperature	1	3	3	0.4047	0.524694		
short_summary	8	50	6	0.9664	0.460282		
long_summary	10	98	10	1.5193	0.125448		
precipIntensity	1	2	2	0.3620	0.547400		

```

precipProbability      1      0      0      0.0002      0.988651
humidity               1     18     18     2.7590     0.096752
windSpeed              1      8      8     1.1795     0.277488
windGust               1      0      0     0.0467     0.828913
visibility             1     10     10     1.5769     0.209252
temperatureHigh        1     10     10     1.5234     0.217141
temperatureLow         1     11     11     1.7563     0.185134
apparentTemperatureHigh 1      0      0     0.0039     0.950027
apparentTemperatureLow  1      1      1     0.0906     0.763397
icon                   2     16      8     1.2100     0.298266
dewPoint               1      2      2     0.2517     0.615874
pressure               1      8      8     1.2796     0.258002
windBearing            1      5      5     0.7611     0.383007
cloudCover             1      4      4     0.5643     0.452540
uvIndex                1      6      6     0.8897     0.345592
ozone                  1      7      7     1.1390     0.285890
sunriseTime           1      1      1     0.1579     0.691133
moonPhase              1      8      8     1.1797     0.277450
precipIntensityMax     1      0      0     0.0007     0.979041
temperatureMin         1     17     17     2.6733     0.102087
temperatureMax         1      0      0     0.0266     0.870428
apparentTemperatureMin  1      1      1     0.2267     0.634005
apparentTemperatureMax  1      4      4     0.5814     0.445779
Residuals              7351  47342      6
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model Summary
S      R-sq    R-sq(adj)    R-sq(pred)
2.537757 92.61%   92.53%      92.78%

```

Note: the following output is missing `vif_manual()` and `summary()`, please check the R-code output.

Overall: The model has strong predictive power and accuracy, but it is misled by the overly complex model with several insignificant and multicollinear predictors. They must be resolved before we finalize our last model, hence, they will be removed henceforth starting from the first transformed model.

To evaluate our selected model, we will perform several hypothesis tests:

Parameter Significance: From TABLE 3.1, we can see that each predictor has a p-value. Let's say that our hypothesis is the following:

$H_0: \beta_i$ is insignificant ($i = 1 \dots k$)

$H_a: \beta_i$ is significant ($i = 1 \dots k$)

We can see that for certain β_i (i.e. *distance*, *name*, *source*, *surge_multiplier*, *days*, etc) that the p-value $< \alpha = 0.05$, however for others, we can see that it is not. What we will do is:

- **Decision:** Reject H_0 , there is enough evidence to say that certain β_i are significant, and others are not.
- **Conclusion:** What we can do here is drop those that are p-value $> \alpha = 0.05$, as they are insignificant, and keep those that are significant (which we will leave for the transformation section)

TABLE 3.2: Summary Statistic Glimpse For Model Adequacy, and Lack of Fit Test

```
> summary(model_basic)
Residual standard error: 2.538 on 7351 degrees of freedom
Multiple R-squared:  0.9261,    Adjusted R-squared:  0.9253 
F-statistic: 1097 on 84 and 7351 DF,  p-value: < 2.2e-16

> print(lack_of_fit_test)
Analysis of Variance Table

  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1    7406 530921
2    7351 47342 55   483579 1365.2 < 2.2e-16 ***
```

Model Adequacy: We want to evaluate the overall performance of the model. From **TABLE 3.2**, we can see the p-value of the overall model. Let's say that our hypothesis is the following:

Ho: The model is inadequate

Ha: The model is adequate

We can see that the p-value of the overall model is around 0 (specifically $2.2e-16$) $< \alpha = 0.05$, therefore:

- **Decision:** Reject Ho, there is enough evidence to say that the model is adequate, at $\alpha = 0.05$
- **Conclusion:** Therefore, from the model statistics, we can say that the model is adequate as well.

Lack-of-Fit Test: Lack-of-fit is often conducted when we have several duplicate values, which applies to our dataset as demonstrated in the EDA. Let's say that our hypothesis is the following:

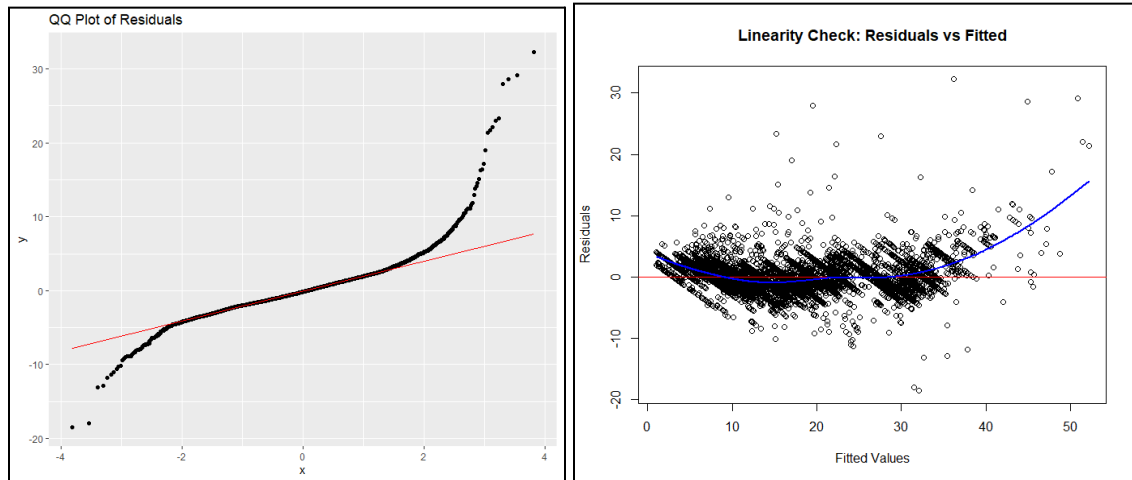
Ho: there is no lack of fit in the model

Ha: there is a lack of fit in the model

From **TABLE 3.2**, the p-value of the lack of fit test demonstrates that the p-value is around 0 $< \alpha = 0.05$, therefore:

- **Decision:** Reject Ho, there is enough evidence to say that the model lacks a lack of fit
- **Conclusion:** This decision demonstrates that the full model has more lack of fit, and therefore error by lack of fit than the reduced model, which consists of the most significant predictors, hence we will simply need to transform, and feature select our data until we can get a model that does not have a lack of fit.

Conclusion: Our model yields that some predictors are significant and that the overall model is adequate, however, it demonstrates a significant lack of fit, therefore, we will aim for transformations, adding interacting terms or feature selection to improve the model's lack of fit. Moreover, severe feature selection is further needed as there are a significant number of insignificant predictors, which unnecessarily increase model complexity, and mask the model's false predictive power through overfitted complexity. Next, we will perform the five linear regression assumptions.

**TABLE 3.3:** Five Assumptions for the Initial Model

---- NORMALITY CHECK ---- Ho: Residuals are normal Ha: Residuals are NOT normal Skewness: 1.731554 Anderson-Darling Test: p-value = 3.7e-24 Decision: Normality MAY BE violated Kolmogorov-Smirnov Test: p-value = 1.170307e-26 Decision: Normality MAY BE violated Normal Probability Plot Correlation: 0.9423562 Decision: Normality MAY BE violated ---- LINEARITY CHECK ---- Ho: The model is linear. Ha: The model is NOT linear. Curvature of loess fit: 1.040948 Decision: Linearity MAY BE violated (loess shows curvature) ---- INDEPENDENCE CHECK ---- Ho: No autocorrelation (independent residuals) Ha: Autocorrelation present Durbin-Watson Test: p-value = 0.504 Decision: Independence NOT violated	---- CONSTANT VARIANCE CHECK ---- Ho: The variances are equal across the groups. Ha: The variances are not equal across the groups. Brown-Forsythe Test: p-value = 9.806497e-53 Decision: Constant variance MAY BE violated (Brown-Forsythe) Ho: The two groups have equal variances. Ha: The two groups have different variances. Variance Test (Assuming Normal Residuals): p-value = 0 Decision: Constant variance MAY BE violated (Variance Test) Levene's Test: p-value = 1.481965e-46 Decision: Constant variance MAY BE violated (Levene's Test) ---- EQUAL MEAN CHECK ---- H0: Mean of residuals = 0 Ha: Mean of residuals $\neq$ 0 Mean of residuals: 1.657946e-18 Decision: Equal mean NOT violated t-Test for Equal Means: p-value = 2.323819e-05 Decision: Equal mean MAY BE violated
--	--

TABLE 3.4: Summary of Linear Regression Assumptions

Violated	Explanation
----------	-------------

Assumptions	
Normality [VIOLATED]	One of the three tests performed, namely the Normal Probability Plot Correlation Test, yields that the residuals are normally distributed, In contrast, KS and AD Tests yield that they are NOT normally distributed. The KS and AD Tests must be taken with a grain of salt, however, as they often are sensitive to large sample sizes of data. That said, from the graph, we can see that while it is normally distributed along the center, it has very heavy skewed tails, which means that overall, it is not normally distributed for the most part.
Constant Variance [VIOLATED]	The residual vs. fitted values graph yields one that does not have constant variance, with an inconsistent fanning shape. Moreover, there is a clear pattern to the variance. This is further supported by the four tests done (Breush-Pagan, Brown-Forsythe, Bartlett Test, and Variance Test), all of which state that the variance is violated. As such, it is clear we will need to apply remedial measures. An important point to note is that the constant variance may be a symptom of the categorical variables. When looking at their scatterplots, it can be seen that some variables, such as name or source, create a vertical series of points, which can have an effect on the residual vs fitted values graph
Equal Mean [NOT VIOLATED]	The graph for the residual vs. fitted values can also be used to determine an equal mean. From a general outlook, we can see, for the most part, that it is equally distributed. This agrees with one of the two tests done for equal mean, where the mean of residuals states that the equal mean is not violated, while the t-test seems to state that it is violated
Independence [NOT VIOLATED]	The graph for residual vs. fitted values can be used to determine the independence. However, it is difficult to really tell if independence is violated. There seem to be patterns by certain predictors, however, we can perform the Durbin-Watson Test to support our assumption: at a significance level of 5%, we can say that we Reject H_0 , as it seems that the independence assumption is violated.
Linearity [VIOLATED]	The residual vs. fitted values graph demonstrates a somewhat minimal polynomial or non-linear relationship. This shows that the linearity assumption is also violated. This is backed up by doing a linearity check, using LOESS fit to check for the linearity assumption violation, which is violated according to the test.

Violated Assumptions & Remedial Measures: As shown by the graphs, tests, and TABLE 3.2, % assumptions are violated, namely normality, constant variance, and linearity. These are significant violations that must be addressed immediately to the best of our abilities. For normality, we will try to $\log(Y)$ or $\sqrt{Y}$, for constant variance we will try to $\log(Y)$ or $\log(Y+k)$, or potentially using Weight Ordinary Least Squares (WLS) which may seem to be the case due to the severity of the violation. Lastly, based on the prototype regression pattern highlighted in the handout, we are facing a vaguely non-linear relationship where the function is opening upwards, hence, we will use $X=X^2$ or $X=\exp(X)$. Box-Cox Transformation will be used first, and then later,

more manual transformations will be employed based on the assumption of violations. Moreover, outliers removal will also be performed, however, instead of the standard $1.5 * IQR$ procedure, we may opt for a more aggressive outlier removal procedure for the sake of fixing the normality assumption (i.e. 1.15 or 1.0). This will come at a severe risk, as too much outlier removal may cause the data to lose important patterns. Interacting or non-linear variables will also be employed; the non-linear assumption violation may indicate that polynomial regression, spline regression, or interacting variables might be needed to solve this problem. Note that while transformations can be used, data-based cannot be resolved, which may be a trait of the data; remedial measures may not help in that case.

3.1.6 Transformation Phase #1: Box-Cox, Weighted Least Squares, and Outlier Removal

This section consists of remedies primarily focused on normality and weighted least squares without resorting to non-linear regression models.

Outlier Removal: Prefacing the two models that will be discussed in this section, from the qqplot of the previous model, we can see some clear outliers. As such, we will employ IQR based on a multiplicity of 1.5 severity as our initial tool, however, should we find more promising results, we may increase the severity. Note that the removal of outliers will not solve our problem, however, if we aim to employ box-cox, we will need to remove them as box-cox is immensely affected by outliers.

Multicollinearity: As mentioned in the initial model analysis, we noticed multicollinearity strongly featured, hence we will employ Manual Post-Analysis Feature Selection. Using the correlation coefficient heatmap, as well as the `vif_manual()` function, we employ critical thinking to remove the features we believe are causing multicollinearity:

- **Similar Metrics:** The dataset contained many different versions of the same main metric being captured. For example, temperature often had temperatureHigh, temperatureLow, temperatureMax, temperatureMin, apparentTemperatureMin and so on. These caused obvious multicollinearity, and we used this along with the ANOVA table, EDA, and the correlation heatmap to decide which ones to keep, and which ones to remove. Another example was a more subtle one; cab_type and name, two very significant predictors, were correlated because every factor level of “name” was prefixed with the cab_type, creating almost a perfect multicollinearity. These sorts of problems initially inhibited VIF analysis, as the function detected perfect multicollinearity and crashed. TABLE 3.5 highlights all the predictors that were removed for multicollinearity.

TABLE 3.5: Manual Post-Analysis Feature Removal/Selection for Multicollinearity

month,	destination,	cab_type,	temperature,	short_summary,	long_summary,	humidity,
temperatureLow,	dewPoint,	pressure,	uvIndex,	ozone,	sunriseTime,	moonPhase,
apparentTemperatureMax,	apparentTemperatureMin,	temperatureMin,	sunsetTime,	windBearing,		
apparentTemperatureLow,	precipProbability,	precipIntensity,	apparentTemperatureHigh,			
temperatureMax						

Model #1: Box-Cox Transformation: We will be using a function specifically, *boxcox()*, to help us find the best lambda value for the transformation. As seen in the figure below, the best lambda value selected was **0.3030303**. This was then applied to the model.

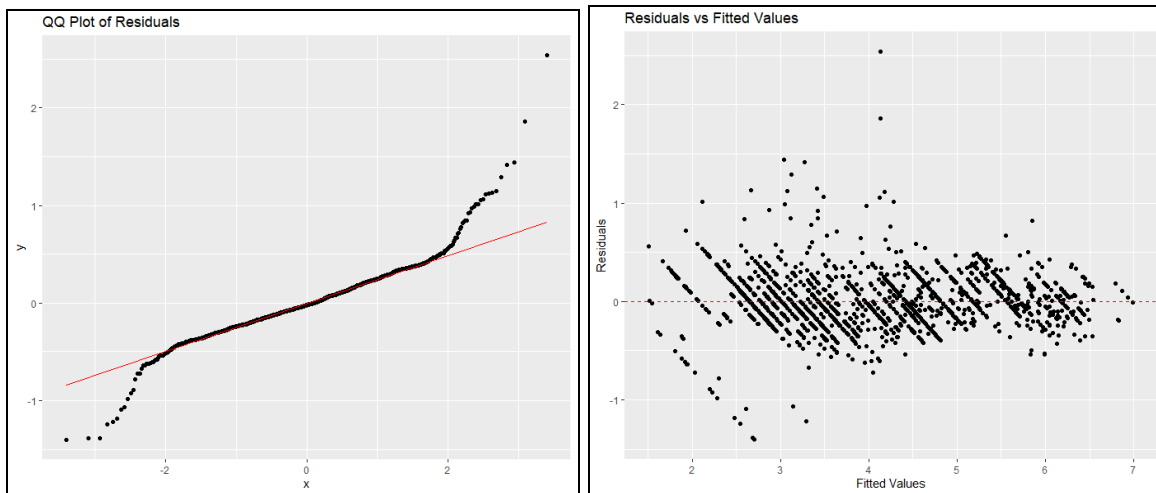
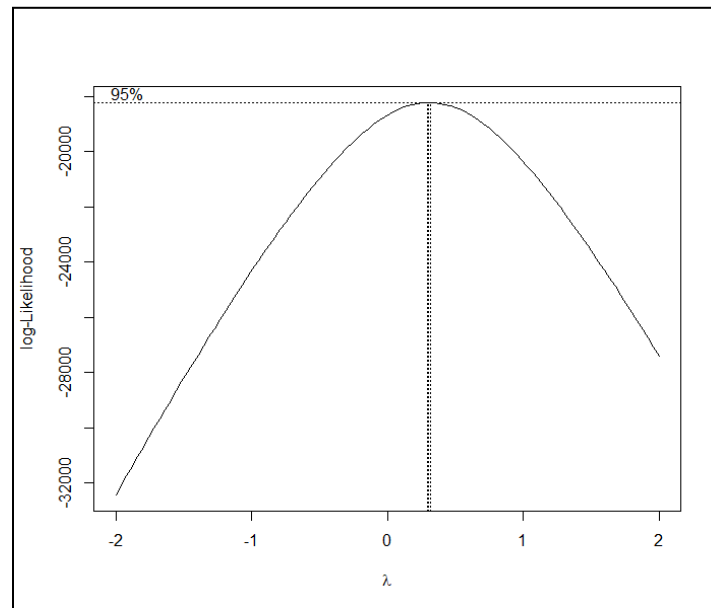


TABLE 3.6: Normality and Constant Variance Assumption

---- NORMALITY CHECK ----	---- CONSTANT VARIANCE CHECK ----
Ho: Residuals are normal Ha: Residuals are NOT normal Skewness: 0.7529613 Anderson-Darling Test: p-value = 3.7e-24 Decision: Normality MAY BE violated	Ho: The variances are equal across the groups. Ha: The variances are not equal across the groups. Brown-Forsythe Test: p-value = 7.322529e-06 Decision: Constant variance MAY BE violated (Brown-Forsythe)

Kolmogorov-Smirnov Test: p-value = 0.0007754469 Decision: Normality MAY BE violated Normal Probability Plot Correlation: 0.9615232 Decision: Normality NOT violated	Ho: The two groups have equal variances. Ha: The two groups have different variances. Variance Test (Assuming Normal Residuals): p-value = 4.679138e-10 Decision: Constant variance MAY BE violated (Variance Test) Levene's Test: p-value = 1.981335e-05 Decision: Constant variance MAY BE violated (Levene's Test)
---	---

- Conclusion:** While the model did improve in terms of predictive power (at 94.94% as shown in the R code), the major improvement was in the normality qqplot shown above. The left tail has almost subsided, however, the right tail is more-or-less similar to the previous qqplot normality distribution, meaning that while it was an improvement, further improvements should be sought. The significance of this should not be undermined, however, as the normal probability plot correlation (TABLE 3.6) had returned that the normality distribution was no longer violated. While KD and AS have not changed, that is to be expected due to the size of the data, though their readings can still provide insight into whether normality is getting better or worse. However, the remaining assumptions remained the same; while box-cox is often powerful, it is susceptible to outliers, which although removed, were still influenced by the non-removed variance data points that are now seen as new outliers after the original outliers were removed.

Model #2: Sqrt(Y) Transformation: As suggested by the handout, we will employ a $\sqrt{Y+k}$ transformation where an appropriate k is selected. The k that was selected was $k=10$, which was done through an observational understanding of the residual qqplots, and experimentation. We aim to solve both normality and constant variance, as they are remedies to both violations.

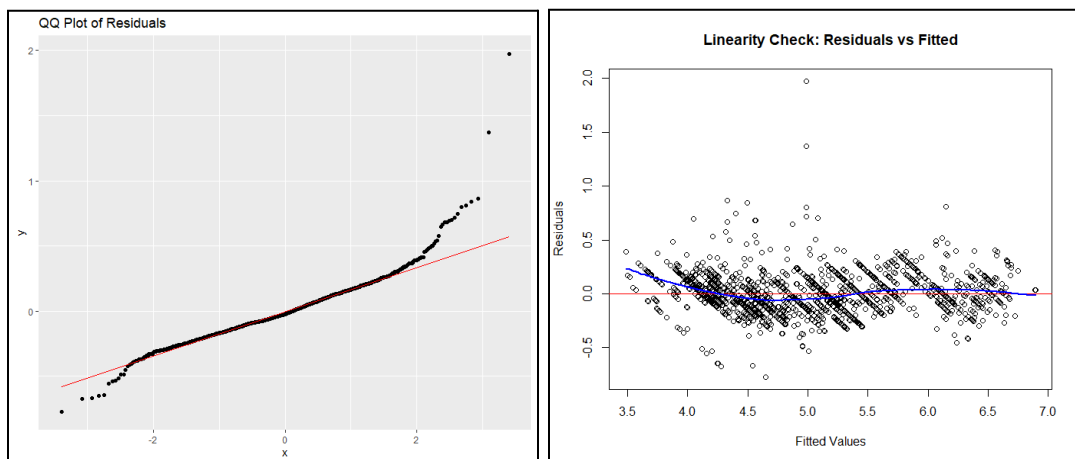
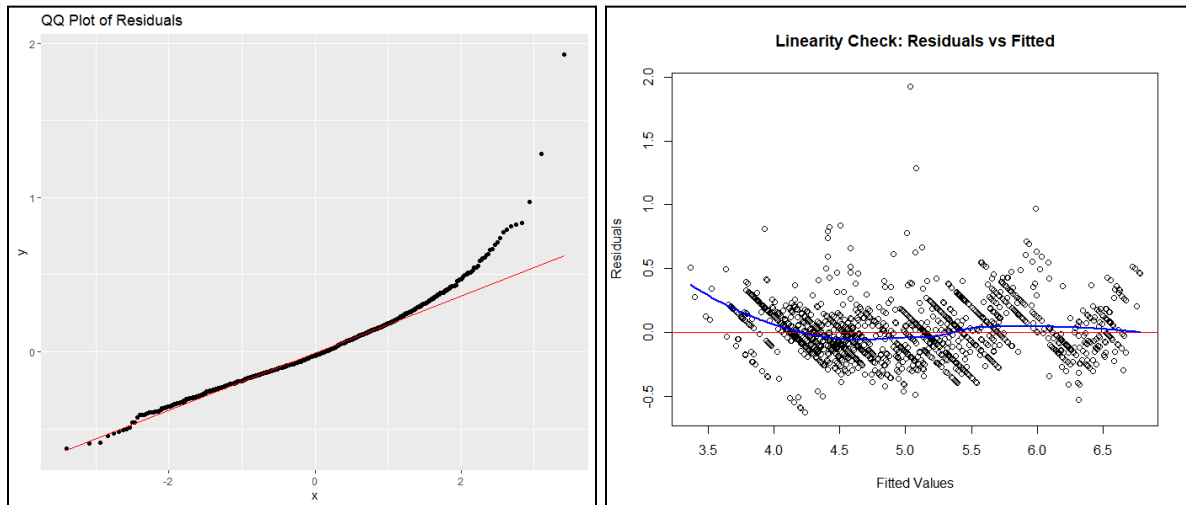


TABLE 3.7: Five Assumptions (Normality and Constant Variance) of Model #2

<pre>---- NORMALITY CHECK ---- Ho: Residuals are normal Ha: Residuals are NOT normal Skewness: 1.343216 Anderson-Darling Test: p-value = 1.970069e-23 Decision: Normality MAY BE violated Kolmogorov-Smirnov Test: p-value = 0.002596853 Decision: Normality MAY BE violated Normal Probability Plot Correlation: 0.9629079 Decision: Normality NOT violated</pre>	<pre>---- CONSTANT VARIANCE CHECK ---- Ho: The variances are equal across the groups. Ha: The variances are not equal across the groups. Brown-Forsythe Test: p-value = 0.09208202 Decision: Constant variance NOT violated (Brown-Forsythe) Ho: The two groups have equal variances. Ha: The two groups have different variances. Variance Test (Assuming Normal Residuals): p-value = 0.006188315 Decision: Constant variance MAY BE violated (Variance Test) Levene's Test: p-value = 0.07377458 Decision: Constant variance NOT violated (Levene's Test)</pre>
--	--

- Conclusion: $\text{Sqrt}(\text{price}+10)$ worked much better than box-cox, perhaps due to the number of outliers making it difficult for box-cox to adjust to. The normality assumption had been officially resolved to a more satisfying degree, with just a slight improvement to the right tail, making it more gradual rather than sudden, and close to the residual normality line. We also see improvements in Brown-Forsythe and Levene's Test, however, no improvement in the F-test. Now that the residuals of the model are normally distributed, the priority should be to make the Variance Test not violate the assumption.

Model #3: Weight Least Squares: Based on the Penn State University description of Weight OLS, it is used by taking each point, and giving them a weightage inversely proportional to the error variance, giving a greater priority to those with smaller error variances. We use weighted OLS, instead of regular OLS, where the weights were set as the inverse of the error variance of the previous model. Moreover, we will also be taking the strongest numerical features we have and transforming them, as a means of attempting to fix the linearity assumption (as per the handout).

**TABLE 3.8:** Five Assumptions (Normality and Constant Variance) of Model #3

---- NORMALITY CHECK ----	---- CONSTANT VARIANCE CHECK ----
Ho: Residuals are normal Ha: Residuals are NOT normal Skewness: 1.215609 Anderson-Darling Test: p-value = 3.7e-24 Decision: Normality MAY BE violated Kolmogorov-Smirnov Test: p-value = 0.0001437903 Decision: Normality MAY BE violated Normal Probability Plot Correlation: 0.9717178 Decision: Normality NOT violate	Ho: The variances are equal across the groups. Ha: The variances are not equal across the groups. Brown-Forsythe Test: p-value = 4.086849e-05 Decision: Constant variance MAY BE violated (Brown-Forsythe) Ho: The two groups have equal variances. Ha: The two groups have different variances. Variance Test (Assuming Normal Residuals): p-value = 1.890504e-08 Decision: Constant variance MAY BE violated (Variance Test) Levene's Test: p-value = 2.046651e-05 Decision: Constant variance MAY BE violated (Levene's Test)

- Conclusion: WLS ended up producing worse results. While it slightly improved normality, model #2 had already resolved the normality assumption violation. Rather, constant variance, now with both Brown-Forsythe and Levene, are both indicating that the assumption is violated. Not to mention that the residual vs fitted graph appears to have had no changes due to the WLS. This could be for a few reasons; for one, the weightage is user-determined, meaning that the weights may not be suitable for the situation. Moreover, the new outliers may still impact the WLS, creating a worse result. Hence, neither WLS (Weighted Least Squares) nor the variance-stabilizing transformations such as $\sqrt{Y+10}$ were able to eliminate the constant variance assumption violation to a satisfactory degree.

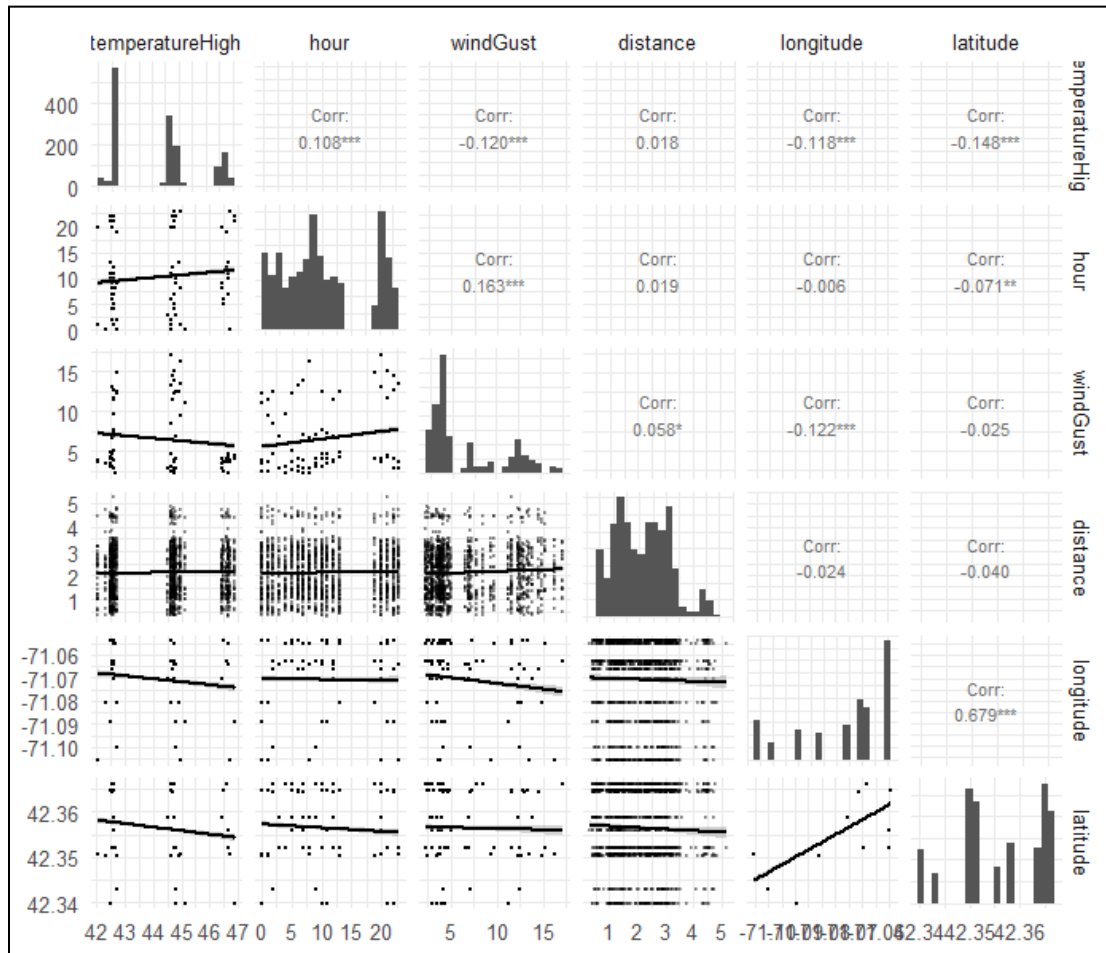
- Moreover, linearity has still been violated slightly, perhaps meaning that the variance and linearity problem coincide. Therefore, we will attempt to see any improvements using polynomial or non-linear regression models.

TABLE 3.8: Five Assumptions (Normality and Constant Variance) of Model #3

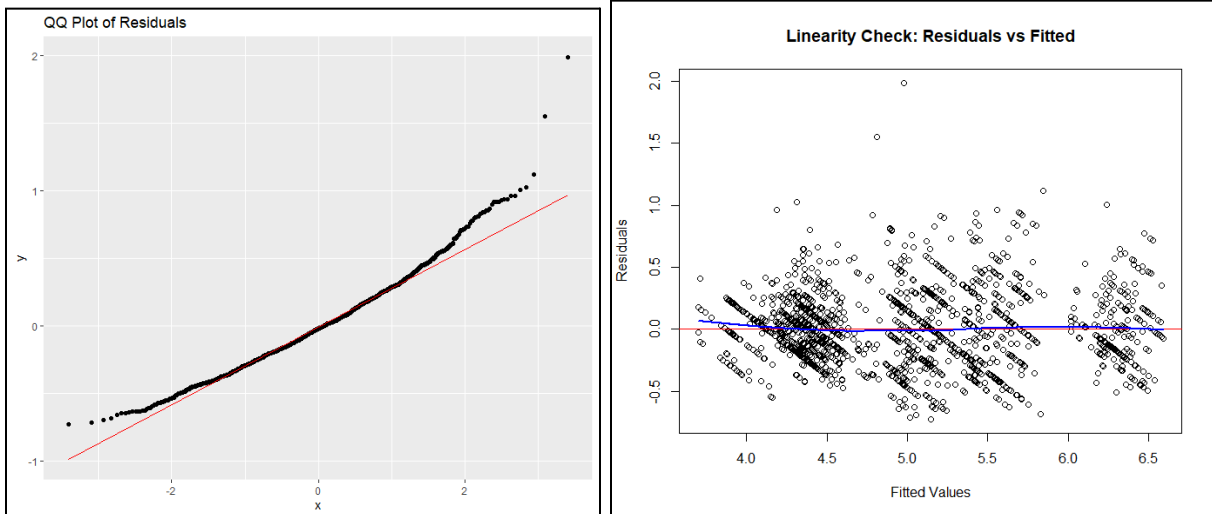
Assumption	Original Assumption (Pre-Phase #1)	Assumption Update (Post-Phase #1)
Normality	[VIOLATED]	[NOT VIOLATED]
Constant Variance	[VIOLATED]	[VIOLATED]
Equal Mean	[NOT VIOLATED]	[NOT VIOLATED]
Independence	[NOT VIOLATED]	[NOT VIOLATED]
Linearity	[VIOLATED]	[VIOLATED]

3.1.7 Transformation Phase #2: Polynomial, Spline, and Generalized Additive Model (GAMs)

Model #4: Polynomial Regression with Spline & Interacting Terms: While the previous section fixed the normality assumption, we still have the constant variance and linearity assumption violated. Often, a violation in linearity stems from a potential requirement in higher-order terms or interaction effects to better capture the underlying relationship. Hence, we will try to use polynomial spline multiple regression models, and Generalized Additive Models (GAMs) to capture any potential non-linear relationships. We will first consult the two-way ANOVA explorations, as well as the correlation coefficient matrix with some of the strongest predictors we have, to see which ones could have interacting relationships, higher-order relationships.



- Based on the two-way ANOVA tables and the aforementioned graph, we can derive several potential relationships and interactions between two variables that are closely associated; we do not want perfect correlation, but rather we would like insight into how variable interactions can provide more data by being stand-alone. In particular, we selected: TemperatureHigh^2 , $\text{WindGust}^2 * \text{Hour}^2$, $\text{TemperatureHigh}^2 * \text{Hour}^2$, $\text{Latitude}^2 * \text{Longitude}^2$. The selection of the following interacting predictors was a mix of exploratory data analysis, observation, and experimentation. Again, our goal with the non-linear regression model is to be able to use the non-linearity as a means to counteracting the linearity assumption violation. Moreover, we kept the $\sqrt{\text{price}+10}$ transformation, as it was the most effective remedies for the normality violation.

**TABLE 3.9:** Five Assumptions (Normality, Constant Variance & Linearity) of Model #4

---- NORMALITY CHECK	CONSTANT VARIANCE CHECK	---- LINEARITY CHECK
Ho: Residuals are normal Ha: Residuals are NOT normal Skewness: 0.7217118 Anderson-Darling Test: p-value = 3.162584e-13 Decision: Normality MAY BE violated Kolmogorov-Smirnov Test: p-value = 0.002169706 Decision: Normality MAY BE violated Normal Probability Plot Correlation: 0.9869217 Decision: Normality NOT violated	Ho: The variances are equal across the groups. Ha: The variances are not equal across the groups. Brown-Forsythe Test: p-value = 1.656549e-18 Decision: Constant variance MAY BE violated (Brown-Forsythe) Ho: The two groups have equal variances. Ha: The two groups have different variances. Variance Test (Assuming Normal Residuals): p-value = 0 Decision: Constant variance MAY BE violated (Variance Test) Levene's Test: p-value = 2.037087e-17 Decision: Constant variance MAY BE violated (Levene's Test)	Ho: The model is linear. Ha: The model is NOT linear. The curvature of loess fit: 0.01423537 Decision: Linearity NOT violated (loess is mostly flat)

Conclusion: All the assumptions remain the same as they had been previously *except* for linearity, which now, according to the strict loess curvature, is now considered a linear non-violation. Meaning out of 5 assumptions, we have successfully satisfied four of them. This comes at a strict price though, as the R^2 predicted value is now

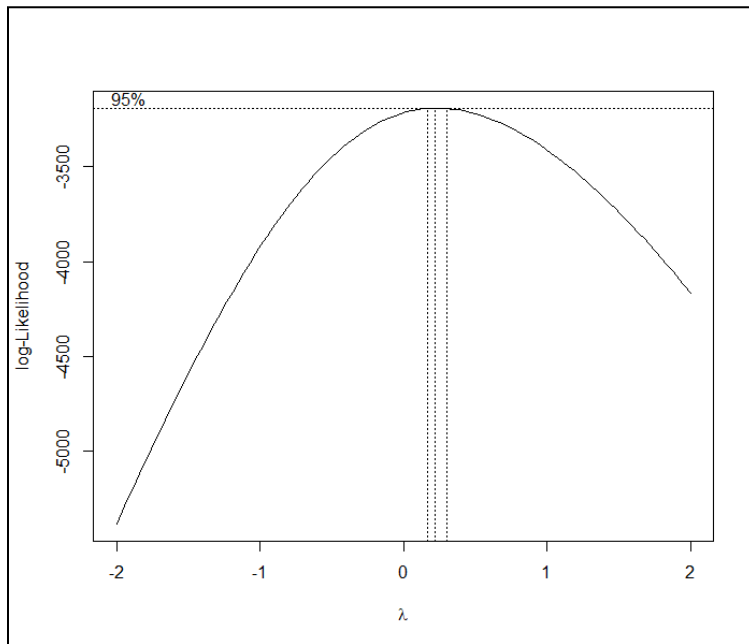
85.48%. This was not directly because of the transition to non-linear, rather the distance metric was causing the linearity assumption to be violated, which provides interesting insight into the feature itself. Finding the right combination for predictor degrees and interacting terms for the polynomial regression is not easy, especially with the number of feature and feature combinations possible. However, we can see its benefit when it comes to the linearity assumption. Moreover, the addition of splines to the non-linear model was to assist with flexibility and to avoid overfitting. Furthermore, despite the interaction, when calling `vif_manual(poly_model)`, it can be seen that all but two parameters violated $VIF < 10$, and even the ones that did, only violated it by 1 point, meaning that overall, despite having a non-linear model, the data does not have substantial multicollinearity. The downside is that despite all the attempts at transformation to fix the constant variance assumption, it remains non-constant.

Model #5: Outlier Removal, Box-Cox, WLS, and Spline-Based Polynomial Regression: As a culmination of all the models studied thus far, we will be making a polynomial regression model with the same parameters as model #4, however, using the WLS weightage (with slight modifications) for from model #3, and using box-cox and outlier removal from model #1. Box-cox with strong outlier removal should be able to help with the normality assumption, WLS with log-based weights should help with constant variance, and linearity should be handled by the polynomial regression model that we are keeping the same. Equal mean and independency have not been violated yet, as such, they will be observed, but require no tuning.

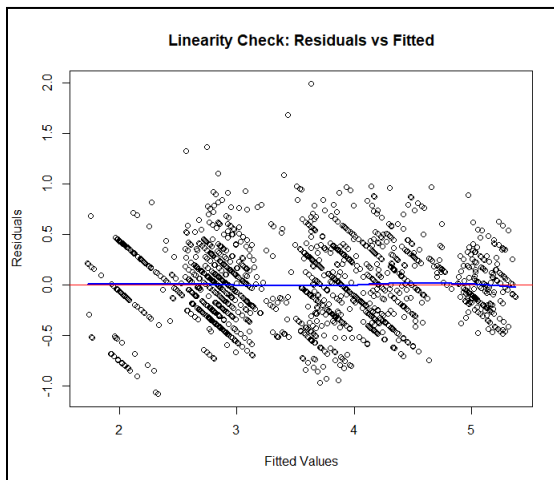
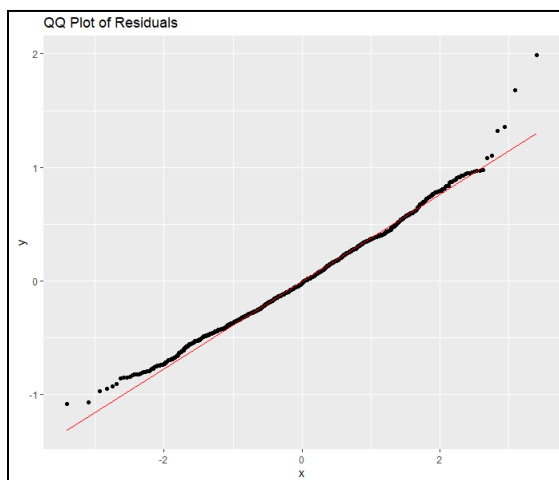
TABLE 3.10: Five Assumptions (Normality and Constant Variance) of Model #5

---- NORMALITY CHECK ----	---- CONSTANT VARIANCE CHECK ----
Ho: Residuals are normal Ha: Residuals are NOT normal Skewness: 0.3289912 Anderson-Darling Test: p-value = 0.001175984 Decision: Normality MAY BE violated Kolmogorov-Smirnov Test: p-value = 0.179015 Decision: Normality NOT violated Normal Probability Plot Correlation: 0.995955 Decision: Normality NOT violated	Ho: The variances are equal across the groups. Ha: The variances are not equal across the groups. Brown-Forsythe Test: p-value = 0.2902999 Decision: Constant variance NOT violated (Brown-Forsythe) Ho: The two groups have equal variances. Ha: The two groups have different variances. Variance Test (Assuming Normal Residuals): p-value = 0.07986786 Decision: Constant variance NOT violated (Variance Test) Levene's Test: p-value = 0.274275 Decision: Constant variance NOT violated (Levene's Test)
---- EQUAL MEAN CHECK ----	---- INDEPENDENCE CHECK ----
H0: Mean of residuals = 0 Ha: Mean of residuals $\neq$ 0 Mean of residuals: 5.731479e-05 Decision: Equal mean MAY BE violated t-Test for Equal Means: p-value =	Ho: No autocorrelation (independent residuals) Ha: Autocorrelation present Durbin-Watson Test: p-value = 0.878 Decision: Independence NOT violated

0.5439469 Decision: Equal mean NOT violated	---- LINEARITY CHECK ---- Ho: The model is linear. Ha: The model is NOT linear. Curvature of loess fit: 0.008978927 Decision: Linearity NOT violated (loess is mostly flat
--	--



- Represents $\lambda = 0.222222\dots$



Model #5 has successfully managed (after using all the previous models' strengths) to fix the violations of all 5 assumptions. This comes at a steep cost though, as the R^2 adjusted is 0.8552, and 0.8515 respectively.

However, despite the relatively low R^2 , it is still high, signifying that the model accounts for 85.52% of the variance. Moreover, while the residual vs. fitted graph is not a perfect case of constant variance, for this project, I find it applicable, as the dataset was not synthesized as is based in the real world. Along with that, the dataset itself is not perfect, as such it becomes a limitation.

Assumption	Original Assumption (Pre-Phase #1)	Assumption Update (Post-Phase #1)	Assumption Update (Post-Phase #2)
Normality	[VIOLATED]	[NOT VIOLATED]	[NOT VIOLATED]
Constant Variance	[VIOLATED]	[VIOLATED]	[NOT VIOLATED]
Equal Mean	[NOT VIOLATED]	[NOT VIOLATED]	[NOT VIOLATED]
Independence	[NOT VIOLATED]	[NOT VIOLATED]	[NOT VIOLATED]
Linearity	[VIOLATED]	[VIOLATED]	[NOT VIOLATED]

3.1.8 Feature Selection: Step-Wise Regression

Feature Selection and Dimensionality Reduction are important aspects of general model building, as we often deal with a large number of features that can easily overwhelm the algorithm producing incorrect results. Due to the sheer number of features. On the other hand, feature selection especially when it comes to interacting variables, can lead to losing significant features that may have been insignificant stand-alone, however, when they are related, they become significant. As such, we will be using stepwise regression, as it is easy to use, and non-exhaustive, which, with large feature spaces, may be computationally expensive. We will also see if our model that abides by 4/5 of the assumptions may be improved or regressed.

3.1.9 Evaluation Metrics, Discussion & Conclusion

Model No.	R^2	R^2 -adj	R^2 -pred	MAE	MAPE	RMSE
0	0.9261	0.9253	0.9278	1.7804	13.3788	2.532
1	0.9421	0.9406	0.9535	1.2567	9.6203	1.9102
2	0.9403	0.9387	0.9463	1.4047	11.015	2.0522
3	0.9373	0.9357	0.9407	1.4454	11.0156	2.1577
4	0.8535	0.48496	0.8548	2.3919	17.2805	3.3754
5	0.8554	0.8515	0.8538	2.3743	16.9669	3.3871

Model No.	Normality	Constant	Equal Mean	Independence	Linear
-----------	-----------	----------	------------	--------------	--------

		Variance			
0	No	No	Yes	Yes	No
1	Yes	No	Yes	Yes	No
2	Yes	No	Yes	Yes	No
3	Yes	No	Yes	Yes	No
4	Yes	No	Yes	Yes	Yes
5	Yes	Yes	Yes	Yes	Yes

The above two tables represent the evaluation metrics per 6 models discussed: R^2 , R^2 -adjusted, R^2 -predicted, MAE, MAPE, and RSME. The next table shows the five model assumptions compared through the 6 models discussed in the paper.

- We can see from the statistics, that the best model based on predictive power, was Model #1 using box-cox and outlier removal.
- However, the model that satisfies all five assumptions is Model #5, with a respectable R^2 of 0.8538, which may not be as impressive as Model #1's 0.9535, but it is still a good score for R^2 .
- Keep in mind, that many of the models still have complex models due to insignificant feature selection. Hence, we will reproduce the score for the best model in terms of predictive power (i.e. Model #1) and the best in terms of assumption validation (Model #5) after performing stepwise regression feature selection.

	R^2	R^2 -adj	R^2 -pred	MAE	MAPE	RMSE
Model #1	0.9417	0.9408	0.9536	1.2595	9.6371	1.9075
Model #5	0.8536	0.8514	0.8525	2.39	17.0933	3.402

	Normality	Constant Variance	Equal Mean	Independence	Linear
Model #1	Yes	No	Yes	Yes	No
Model #5	Yes	Yes	Yes	Yes	Yes

- Overall, we managed to find two excellent models in their own category: one for predictive power, and another one for statistical inference such as confidence and prediction intervals

3.1.10 Confidence Interval & Prediction Interval

One application of the model is to use it for confidence and prediction intervals. For the sake of proper inference, we will use Model #5, one that satisfies (to the best of our abilities), the 5 model assumptions, hence allowing for more accurate inference over the other models (i.e. Model #1).

Section 3.2: Classification Model Through Naive Bayes & Logistic Regression

Naive Bayes Classification:

(In this section the original sampling of $X = 250$ was used.)

For further classification, assuming independence, we used Naive Bayes Classification. In this section we made sure to use independent variables that we got from previous EDA tests. Naive bayes classification was applied to predict four target variables: cab_type, name, source and short_summary, using both unbinned (continuous values) and binned versions of the dataset. Additionally to further validate the model, we applied K-folding cross validation of 5 folds to our model. The Naive Bayes classification models produced mixed results depending on the target variable and whether the data was binned or not.

Model 1: (Predicting cab_type)

Unbinned: The first model achieved relatively strong training accuracy of 54.43%, and its 5-fold cross-validation accuracy was 54.42%, indicating a highly consistent performance with low overfitting. This suggests that the predictors (hour, source, distance, and surge_multiplier) provide reasonable information about the type of cab (uber vs lyft).

Binned: The first Binned Model achieved 55.05% training accuracy and 51.60% cross validation (CV) accuracy, showing that binning preserved most of the model's predictive strength while potentially improving robustness.

Model 2: (Predicting name)

Unbinned: The second model achieved 51.59% accuracy on the training set but dropped to 45.62% in CV, suggesting mild overfitting. The model captures some predictive patterns from price, distance, and other features, but may be sensitive to class imbalance or noise.

Binned: The binned second model's training accuracy dropped to 34.04% and just 20.97% in CV, which is significantly worse than the unbinned version. This suggests binning may have removed important data from price and distance, which are important to identifying specific cab names.

Model 3: (Predicting source)

Unbinned: The third model performed poorly, with just 8.87% training accuracy and 8.68% in CV. This consistently low performance suggests that the input features (hour, cab_type, surge_multiplier, temperature, windBearing) are not sufficient to predict pickup source accurately, possibly due to limited correlation or overlapping distributions between classes.

Binned: The third Binned Model showed little change, with 9.48% training and 8.54% CV accuracy, which is still far below acceptable accuracy. This again reflects that the chosen features are not well-suited to predicting source, and binning neither helped nor hurt meaningfully.

Model 4: (Predicting short_summary)

Unbinned: The fourth model reached 42.12% training accuracy but suffered a sharp drop to 3.38% in CV. This highlights clear overfitting, likely due to the model memorizing noise in the training data without learning patterns that generalize. The target variable might also have too many unique values for Naive Bayes to handle effectively with the given features.

Binned: The fourth Binned Model performed similarly poorly to the unbinned model, with a negligible improvement in CV accuracy (3.63% vs. 3.38%), indicating that the model cannot generalize patterns for weather summaries from the given features.

Confusion Matrix of Model 1:

<i>Unbinned Confusion Matrix:</i>	<i>Binned Confusion Matrix:</i>
Actual Predicted Lyft Uber Lyft 2068 0 Uber 37637 42892 Model 1 Accuracy: 0.5443297 --> 54.43 %	Actual Predicted Lyft Uber Lyft 7147 4569 Uber 32558 38323 Model 1 Accuracy: 0.5505043 --> 55.05 %
<i>Unbinned 5-fold CV:</i>	<i>Binned 5-fold CV:</i>
CV Unbinned Model 1 Accuracy: 54.42328 %	CV Binned Model 1 Accuracy: 51.59873 %

The best-performing and most reliable model was Model 1 (Predicting cab_type), showing consistent results across training, cross-validation, and both data formats. This makes it a suitable candidate for real-world application in distinguishing between ride types. Models 3 and 4 were the weakest performers, showing extremely low accuracy and significant overfitting. Additionally Binning proved helpful in some contexts (Model 1) but harmful in others (Model 2), especially when numeric precision was important. Overall, these results suggest that while Naive Bayes can be effective for some classification tasks, its assumptions (such as feature independence) and simplicity limit its effectiveness for complex or noisy targets.

Logistic Regression:

(In this section the original sampling of $X = 250$ was used.)

Model 1: Predicting Cab Type (Uber vs Lyft)

In this model, we used a binary logistic regression to predict the probability of a cab being of a particular type (cab_type) (e.g., Uber vs. Lyft) using predictors such as hour, day, distance, price, temperature, and humidity. To prepare the variables we converted all categorical variables such as (cab_type, hour and day) to factors. Additionally we omitted missing values and made cab_type our target variable as it is binary and suitable for logistic regression.

Coefficients:				
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.1595155	0.0439546	3.629	0.000284 ***

```

hour1      0.0366624  0.0170579   2.149 0.031612 *
hour2      0.0566842  0.0171246   3.310 0.000933 ***
hour3      0.0482212  0.0172476   2.796 0.005177 **
hour4      0.0505353  0.0171809   2.941 0.003268 **
hour5      0.0253817  0.0178779   1.420 0.155689
hour6      0.0658279  0.0175138   3.759 0.000171 ***
hour7      0.0393775  0.0179755   2.191 0.028479 *
hour8      0.0474042  0.0182360   2.599 0.009336 **
hour9      0.0208004  0.0174197   1.194 0.232449
hour10     0.0526614  0.0172756   3.048 0.002301 **
hour11     0.0505136  0.0171703   2.942 0.003262 **
hour12     0.0159911  0.0170742   0.937 0.348982
hour13     0.0326752  0.0169599   1.927 0.054028 .
hour14     0.0589229  0.0168846   3.490 0.000483 ***
hour15     0.0365796  0.0170928   2.140 0.032350 *
hour16     0.0514751  0.0173300   2.970 0.002975 **
hour17     -0.0135404  0.0175290  -0.772 0.439845
hour18     0.0413879  0.0178748   2.315 0.020589 *
hour19     -0.0022651  0.0186347  -0.122 0.903253
hour20     0.0580001  0.0186506   3.110 0.001872 **
hour21     0.0069189  0.0181668   0.381 0.703312
hour22     0.0070325  0.0175475   0.401 0.688593
hour23     0.0449021  0.0171173   2.623 0.008711 **
day2       0.0115233  0.0163439   0.705 0.480777
day3       -0.0103320  0.0182967  -0.565 0.572285
day4       -0.0253026  0.0234286  -1.080 0.280149
day9       0.2040674  0.0545342   3.742 0.000183 ***
day10      0.0403378  0.0428295   0.942 0.346283
day13      0.0475152  0.0181180   2.623 0.008728 **
day14      -0.0011880  0.0140440  -0.085 0.932586
day15      0.0051885  0.0171465   0.303 0.762194
day16      0.0020446  0.0145877   0.140 0.888532
day17      0.0354995  0.0152516   2.328 0.019934 *
day18      0.0406621  0.0165590   2.456 0.014065 *
day26      0.0051434  0.0180278   0.285 0.775412
day27      0.0108171  0.0145899   0.741 0.458443
day28      0.0157869  0.0127021   1.243 0.213920
day29      0.0277497  0.0136763   2.029 0.042455 *
day30      0.0290086  0.0139722   2.076 0.037878 *
distance    0.0619825  0.0023671  26.185 < 2e-16 ***
price      -0.0206291  0.0002898 -71.186 < 2e-16 ***
temperature 0.0012221  0.0009259   1.320 0.186863
humidity    0.0283998  0.0383046   0.741 0.458438
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 883582 on 637975 degrees of freedom
Residual deviance: 878332 on 637932 degrees of freedom
AIC: 878420

```

Number of Fisher Scoring iterations: 3

Odd Ratios:

Odds Ratios for Cab Type Model:

(Intercept)	hour1	hour2	hour3	hour4
1.1729424	1.0373427	1.0583216	1.0494027	1.0518340
hour5	hour6	hour7	hour8	hour9
1.0257065	1.0680429	1.0401631	1.0485457	1.0210183
hour10	hour11	hour12	hour13	hour14
1.0540726	1.0518112	1.0161197	1.0332149	1.0606935
hour15	hour16	hour17	hour18	hour19
1.0372568	1.0528229	0.9865509	1.0422563	0.9977375
hour20	hour21	hour22	hour23	day2
1.0597151	1.0069429	1.0070572	1.0459255	1.0115900
day3	day4	day9	day10	day13
0.9897212	0.9750149	1.2263808	1.0411624	1.0486621
day14	day15	day16	day17	day18
0.9988127	1.0052020	1.0020467	1.0361371	1.0415002
day26	day27	day28	day29	day30
1.0051566	1.0108759	1.0159122	1.0281383	1.0294335
distance	price	temperature	humidity	
1.0639437	0.9795823	1.0012228	1.0288069	

The significant predictors include:

- Distance was strongly significant ($p < 2e-16$) with a positive coefficient ($\beta = 0.06198$) indicating that longer trips are more likely to be associated with one cab type. The odds ratio = 1.063, meaning each additional unit of distance increases the odds of choosing that cab type by ~6.4%.
- Price was highly significant ($p < 2e-16$) with a negative coefficient ($\beta = -0.0206$) showing that higher prices are associated with a lower probability of choosing that cab type. The odds ratio = 0.979, indicating each dollar increase reduces the odds by ~2.1%.
- Hour of the day was a significant predictor for many time slots ($p < 0.05$) like hour6, hour10, hour14, hour20, suggesting cab type usage varies throughout the day. For example, the odds ratio for hour6 was 1.068, meaning the odds of getting a particular cab type were 6.8% higher at 6 AM compared to the reference hour.
- Most days of the month were not significant, but a few such as day9, day13, and day17 showed higher odds, potentially indicating specific patterns in service demand or promotions.

Odds Ratio CI Interpretation:

Odds ratios were calculated and their 95% confidence intervals were also examined. For instance, day9 had an odds ratio of **1.226** (CI: approx. 1.10–1.37), implying a 22.6% higher likelihood of one cab type being chosen on that day.

Odds Ratio CI for Cab Type (95%):

	2.5 %	97.5 %
(Intercept)	1.0761243	1.2784712
hour1	1.0032347	1.0726104

hour2	1.0233900	1.0944454
hour3	1.0145210	1.0854838
hour4	1.0170044	1.0878564
hour5	0.9903880	1.0622845
hour6	1.0320028	1.1053415
hour7	1.0041548	1.0774626
hour8	1.0117306	1.0867005
hour9	0.9867470	1.0564798
hour10	1.0189797	1.0903742
hour11	1.0170033	1.0878104
hour12	0.9826781	1.0506993
hour13	0.9994345	1.0681370
hour14	1.0261661	1.0963825
hour15	1.0030830	1.0725949
hour16	1.0176632	1.0891975
hour17	0.9532323	1.0210340
hour18	1.0063742	1.0794178
hour19	0.9619542	1.0348519
hour20	1.0216773	1.0991692
hour21	0.9717201	1.0434424
hour22	0.9730108	1.0422950
hour23	1.0114175	1.0816108
day2	0.9796987	1.0445194
day3	0.9548578	1.0258576
day4	0.9312553	1.0208307
day9	1.1020614	1.3647242
day10	0.9573307	1.1323353
day13	1.0120768	1.0865699
day14	0.9716946	1.0266876
day15	0.9719820	1.0395574
day16	0.9738026	1.0311101
day17	1.0056227	1.0675775
day18	1.0082409	1.0758565
day26	0.9702607	1.0413076
day27	0.9823786	1.0401998
day28	0.9909326	1.0415214
day29	1.0009451	1.0560703
day30	1.0016250	1.0580140
distance	1.0590191	1.0688913
price	0.9790260	0.9801388
temperature	0.9994076	1.0030414
humidity	0.9543968	1.1090184

Multicollinearity:

Variance Inflation Factors (VIFs) were also inspected. By checking the table we can see that the VIFs remained within acceptable ranges (< 10), indicating no severe multicollinearity among predictors.

	GVIF	Df	$GVIF^{(1/(2*Df))}$
hour	3.778314	23	1.029319
day	17.534350	16	1.093633
distance	1.137335	1	1.066459

price	1.137188	1	1.066390
temperature	6.125874	1	2.475050
humidity	4.450900	1	2.109716

Model 2: Predicting Surge Pricing (is_surge)

This model aimed to classify whether surge pricing (is_surge) was in effect based on ccab_type, hour, day, distance, temperature, humidity and windSpeed. To prepare the data, we first converted all relevant categorical variables into factor variables, and removed any missing values. The target variable, is_surge is binary (0 = no surge, 1 = surge), making it well-suited for logistic regression. This model allows us to examine how both temporal and environmental factors influence the likelihood of surge pricing being applied.

```

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.662e+00  1.302e-01 -20.440 < 2e-16 ***
cab_typeUber -1.795e+01  3.077e+01  -0.583 0.559643
hour1        -3.017e-02  4.811e-02  -0.627 0.530557
hour2        -5.006e-03  4.802e-02  -0.104 0.916965
hour3         1.161e-02  4.830e-02   0.240 0.809963
hour4        -3.726e-02  4.867e-02  -0.765 0.443997
hour5        -3.510e-02  5.049e-02  -0.695 0.486957
hour6        -2.642e-02  4.972e-02  -0.531 0.595214
hour7        -1.086e-01  5.187e-02  -2.094 0.036238 *
hour8         3.704e-03  5.127e-02   0.072 0.942401
hour9        -4.872e-02  4.935e-02  -0.987 0.323612
hour10       -5.579e-02  4.906e-02  -1.137 0.255415
hour11       -4.558e-02  4.861e-02  -0.938 0.348424
hour12       -8.942e-03  4.773e-02  -0.187 0.851388
hour13        1.494e-01  4.602e-02   3.247 0.001167 **
hour14       -3.640e-02  4.789e-02  -0.760 0.447242
hour15       -6.476e-02  4.904e-02  -1.321 0.186616
hour16       -1.733e-01  5.097e-02  -3.400 0.000674 ***
hour17       -4.881e-02  4.988e-02  -0.978 0.327867
hour18       -2.843e-02  5.094e-02  -0.558 0.576789
hour19       -8.668e-02  5.334e-02  -1.625 0.104169
hour20        6.260e-02  5.205e-02   1.203 0.229095
hour21        1.051e-04  5.107e-02   0.002 0.998358
hour22        1.690e-02  4.900e-02   0.345 0.730268
hour23       -4.844e-02  4.891e-02  -0.990 0.321961
day2         -4.145e-02  4.732e-02  -0.876 0.381045
day3          2.435e-02  5.275e-02   0.462 0.644423
day4         -2.478e-02  6.826e-02  -0.363 0.716550
day9         -2.095e-02  1.606e-01  -0.130 0.896204
day10         1.453e-01  1.128e-01   1.287 0.197941
day13        -6.005e-02  5.199e-02  -1.155 0.248081
day14        -7.590e-02  4.019e-02  -1.888 0.058967 .
day15        -1.693e-04  4.941e-02  -0.003 0.997267
day16         4.334e-02  4.423e-02   0.980 0.327080
day17         8.100e-02  4.924e-02   1.645 0.099925 .
day18         2.221e-02  5.620e-02   0.395 0.692692
day26        -4.388e-02  5.277e-02  -0.831 0.405697

```

```

day27      6.953e-02  5.139e-02  1.353 0.176060
day28      2.195e-02  3.973e-02  0.553 0.580518
day29      3.192e-02  4.607e-02  0.693 0.488445
day30     -3.992e-02  3.979e-02 -1.003 0.315694
distance    1.372e-01  6.388e-03 21.478 < 2e-16 ***
temperature -2.770e-03  2.653e-03 -1.044 0.296362
humidity    -1.332e-01  1.114e-01 -1.195 0.231967
windSpeed   -5.977e-03  4.483e-03 -1.333 0.182411
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 184511  on 637975  degrees of freedom
Residual deviance: 152562  on 637931  degrees of freedom
AIC: 152652

Number of Fisher Scoring iterations: 19

```

Odd Ratios:

Odds Ratios for Surge Model:

```

(Intercept) cab_typeUber      hour1      hour2      hour3
6.980097e-02 1.596596e-08 9.702775e-01 9.950063e-01 1.011682e+00
      hour4      hour5      hour6      hour7      hour8
9.634290e-01 9.655075e-01 9.739280e-01 8.970576e-01 1.003711e+00
      hour9      hour10      hour11      hour12      hour13
9.524520e-01 9.457357e-01 9.554424e-01 9.910978e-01 1.161154e+00
      hour14      hour15      hour16      hour17      hour18
9.642559e-01 9.372887e-01 8.408714e-01 9.523655e-01 9.719732e-01
      hour19      hour20      hour21      hour22      hour23
9.169705e-01 1.064606e+00 1.000105e+00 1.017039e+00 9.527114e-01
      day2      day3      day4      day9      day10
9.593971e-01 1.024645e+00 9.755200e-01 9.792689e-01 1.156331e+00
      day13      day14      day15      day16      day17
9.417162e-01 9.269099e-01 9.998307e-01 1.044295e+00 1.084374e+00
      day18      day26      day27      day28      day29
1.022457e+00 9.570680e-01 1.072006e+00 1.022196e+00 1.032431e+00
      day30      distance      temperature      humidity      windSpeed
9.608630e-01 1.147065e+00 9.972337e-01 8.753180e-01 9.940405e-01

```

Interpretation of Model and ORs:

- Distance was the most significant predictor ($p < 2e-16$), indicating that longer trips were much more likely to have surge pricing, with an odds ratio of 1.15 (95% CI: 1.13–1.16), indicating a 15% increase in odds of surge per unit increase in distance.
- Cab type was not statistically significant in this model ($p = 0.56$). The extremely low odds ratio ($\approx 1.6e-08$) and wide confidence interval suggest high uncertainty, likely due to imbalance or low variability across cab types during surge periods.

- Hour of day showed mixed significance. For example: hour7 was associated with *lower* odds of surge pricing (OR = 0.90, $p = 0.036$), hour13 had *higher* odds (OR = 1.16, $p = 0.0012$), and hour16 significantly reduced the odds (OR = 0.84, $p < 0.001$).
- Environmental factors such as (temperature, humidity, windSpeed) were not statistically significant, but showed some directional influence (e.g., higher humidity slightly reduced the odds of surge).

Odds Ratio Interpretation:

Odds ratios were calculated and their 95% confidence intervals were also examined. For instance, the odds ratio for distance being **1.15** means that with each additional unit of distance, the likelihood of surge pricing increases by **15%**, holding all else constant.

Odds Ratio CI for Surge (95%):

	2.5 %	97.5 %
(Intercept)	5.407560e-02	9.009934e-02
cab_typeUber	1.018985e-34	2.501627e+18
hour1	8.829656e-01	1.066223e+00
hour2	9.056336e-01	1.093199e+00
hour3	9.203076e-01	1.112129e+00
hour4	8.757700e-01	1.059862e+00
hour5	8.745305e-01	1.065949e+00
hour6	8.834910e-01	1.073622e+00
hour7	8.103375e-01	9.930583e-01
hour8	9.077493e-01	1.109818e+00
hour9	8.646350e-01	1.049188e+00
hour10	8.590382e-01	1.041183e+00
hour11	8.686127e-01	1.050952e+00
hour12	9.025865e-01	1.088289e+00
hour13	1.061009e+00	1.270753e+00
hour14	8.778636e-01	1.059150e+00
hour15	8.513951e-01	1.031848e+00
hour16	7.609228e-01	9.292200e-01
hour17	8.636609e-01	1.050181e+00
hour18	8.796230e-01	1.074019e+00
hour19	8.259433e-01	1.018030e+00
hour20	9.613480e-01	1.178954e+00
hour21	9.048474e-01	1.105391e+00
hour22	9.239005e-01	1.119566e+00
hour23	8.656218e-01	1.048563e+00
day2	8.744204e-01	1.052632e+00
day3	9.239980e-01	1.136255e+00
day4	8.533561e-01	1.115172e+00
day9	7.148511e-01	1.341493e+00
day10	9.269314e-01	1.442503e+00
day13	8.504814e-01	1.042738e+00
day14	8.566957e-01	1.002879e+00
day15	9.075390e-01	1.101508e+00
day16	9.575864e-01	1.138855e+00
day17	9.846228e-01	1.194232e+00
day18	9.158216e-01	1.141510e+00
day26	8.630217e-01	1.061363e+00

day27	9.692884e-01	1.185608e+00
day28	9.456263e-01	1.104967e+00
day29	9.432925e-01	1.129993e+00
day30	8.887751e-01	1.038798e+00
distance	1.132793e+00	1.161517e+00
temperature	9.920624e-01	1.002432e+00
humidity	7.036139e-01	1.088923e+00
windSpeed	9.853448e-01	1.002813e+00

Multicollinearity:

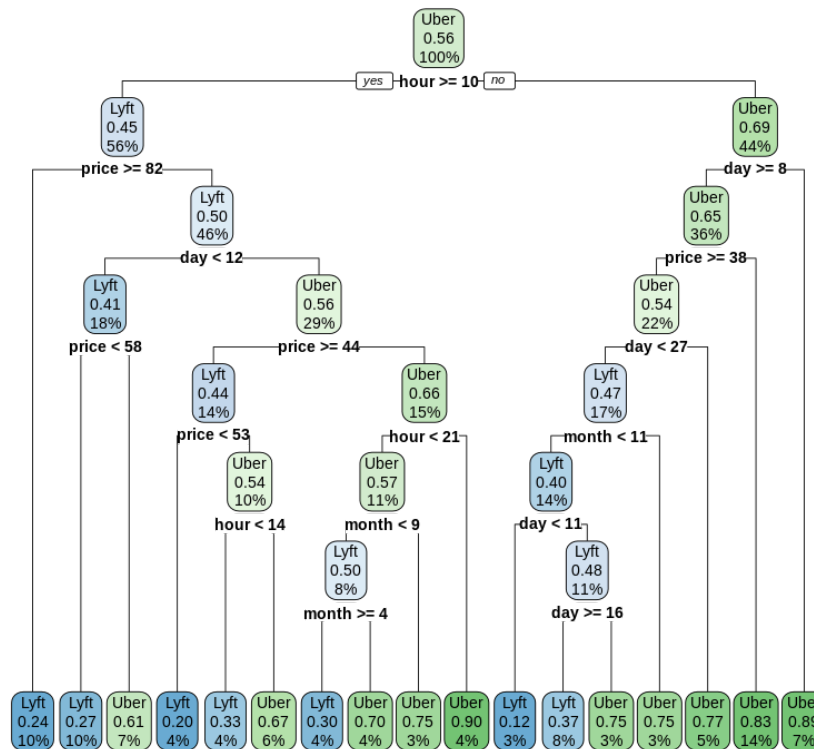
Variance Inflation Factors (VIFs) for this model were also inspected. By checking the table we can see that the VIFs remained within acceptable ranges (< 10), indicating no severe multicollinearity among predictors.

	GVIF	Df	GVIF ^{1/(2*Df)}
cab_type	1.000000	1	1.000000
hour	4.024343	23	1.030731
day	62.978688	16	1.138216
distance	1.000381	1	1.000190
temperature	6.190557	1	2.488083
humidity	4.614347	1	2.148103
windSpeed	3.867211	1	1.966522

Section 3.3: Decision Tree

Decision Tree:

Decision Tree for Cab Type Prediction



The decision tree that was generated provides a clear visual breakdown of how the `surge_multiplier` is predicted based on key factors like the time of day (hour), the day of the month (day), the month itself (month), and the ride price (price). We started by loading the `rideshare_kaggle.csv` dataset and selected key numerical features like hour, day, month, price, distance, latitude, longitude, temperature, and `apparentTemperature`, along with the target variable `surge_multiplier`. After removing missing values and converting the target variable to a categorical factor, the data was split into a training set (70%) and a testing set (30%) to assess the model's performance.

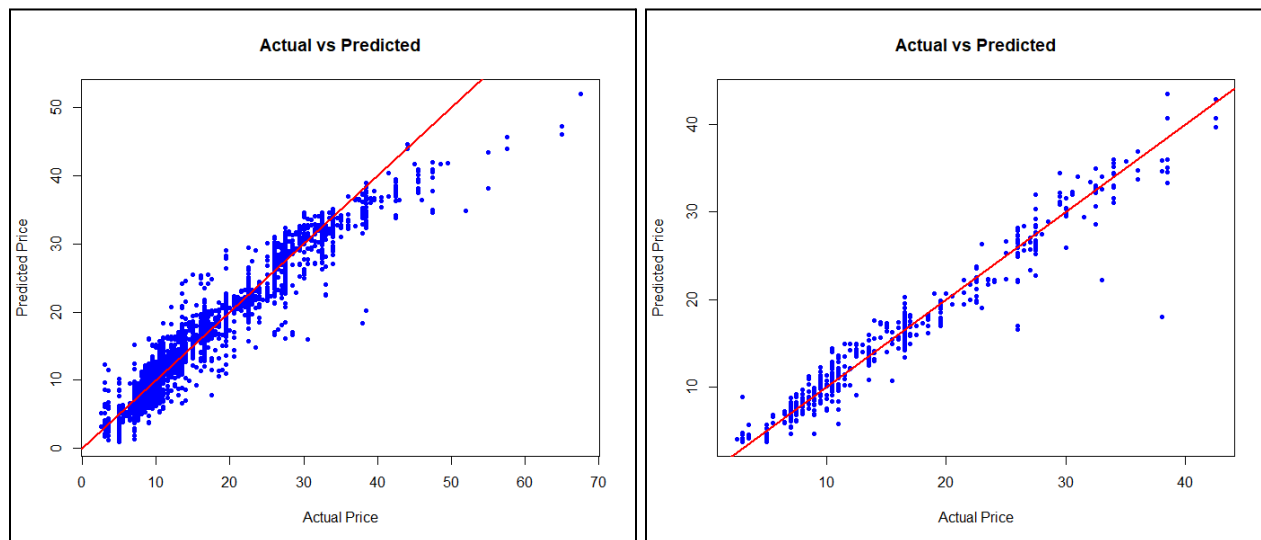
A decision tree model was then built using the `rpart` package with the `class` method to handle the categorical nature of `surge_multiplier`. The tree structure was visualized, making it easier to see how the model makes decisions based on the features. The model's predictions on the test data were then analyzed through a confusion matrix generated using the `confusionMatrix()` function from the `caret` package. This matrix provided a detailed breakdown of how well the model predicted each surge multiplier class, capturing true positives, false positives, true negatives, and false negatives. At the top of

the tree, we see that the first major decision point is the hour. If the ride is booked at or after 10 AM, the model leans toward predicting it as an Uber ride, with a 56% probability of being classified that way. This suggests that Uber rides tend to experience higher surge pricing during later hours, possibly due to increased demand.

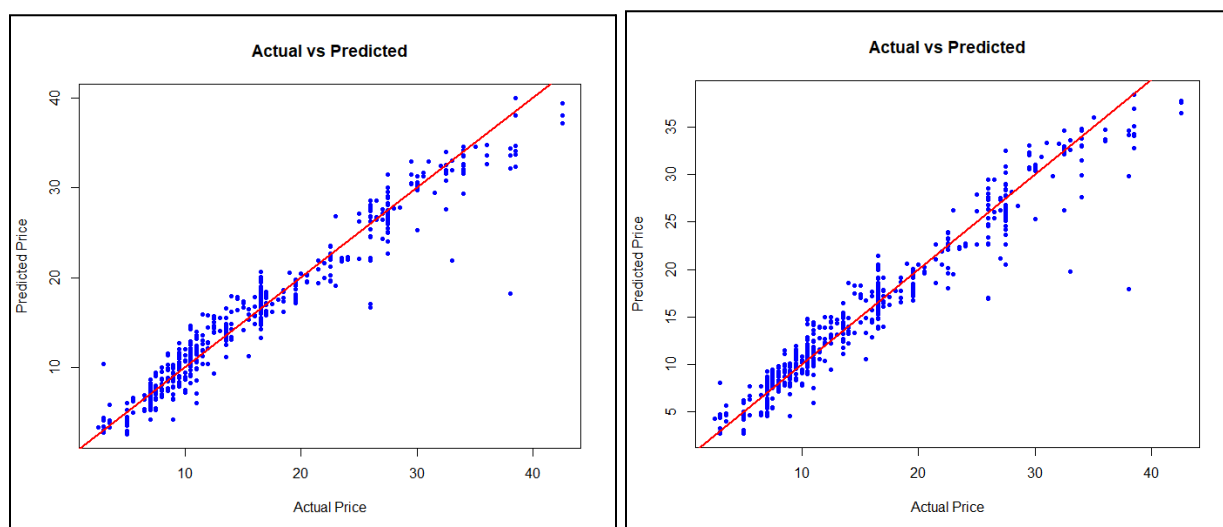
Moving further down the tree, we see how other factors like price and day come into play. For instance, when the price is less than 58 and the day is before the 12th, the model is more likely to predict Lyft, with the probability dropping to 41%. However, when the price is 44 or higher, and the hour is before 9 PM, the model swings back to predicting Uber, indicating that Uber rides are more common or more expensive during those hours. The nodes at the bottom of the tree show the final classifications, with each leaf node representing a specific combination of conditions that lead to either Lyft or Uber. The color coding, blue for Lyft and green for Uber, helps to visually distinguish the outcomes, with the probabilities showing how confident the model is in each prediction.

This tree structure was formed by analyzing the data to identify the most influential splits that reduce classification uncertainty. It effectively captures how different time periods, days of the month, and price ranges impact the likelihood of a ride being classified as a Lyft or Uber, providing a clear, step-by-step breakdown of how surge multipliers are influenced by these factors.

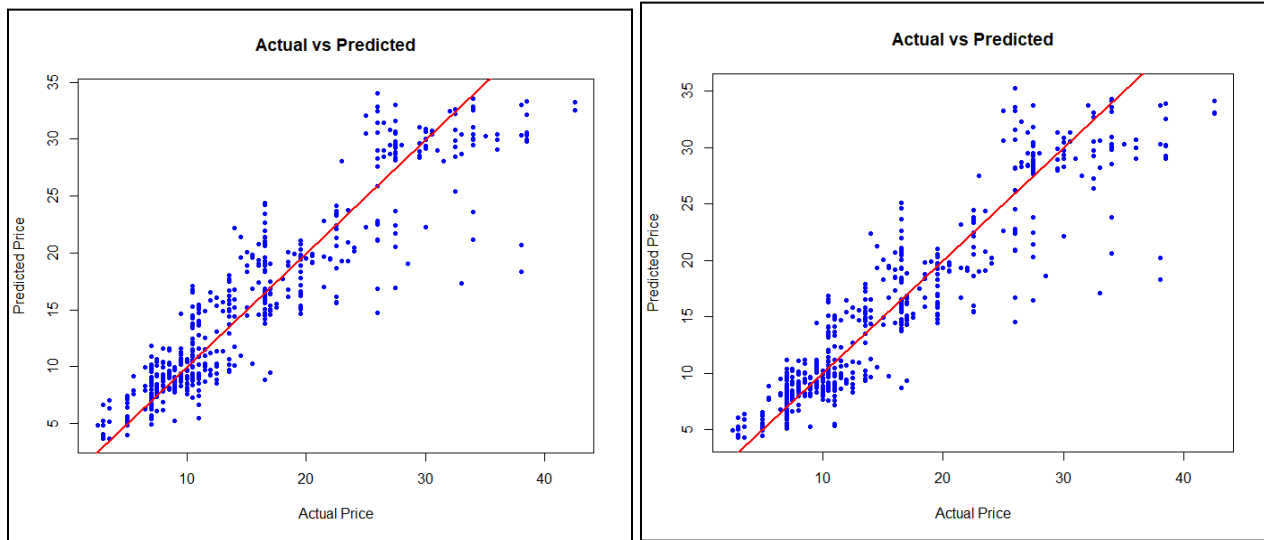
Section 4: Model Validation



Model 0 & Model 1, Accuracy vs. Prediction Graph



Model 2 & Model 3, Accuracy vs. Prediction Graph



Model 4 & Model 5, Accuracy vs. Prediction Graph

The above figures represent the actual vs predicted scatterplot. It allows us to see how accurate a prediction is based against its actual value. If data adheres to the diagonal then that implies it is a good model. This is reflected in the statistics too; as we know about R^2 , R^2 -adjusted and R^2 -predicted, metrics that represent how many a model can account for variance. Overall all models have 85+ R^2 , which is overall, very good. Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error. All metrics used for quality for regression, and can be compared. For example, model #5, has the highest root square mean error.

Model No.	R^2	R^2 -adj	R^2 -pred	MAE	MAPE	RMSE
0	0.9261	0.9253	0.9278	1.7804	13.3788	2.532
1	0.9421	0.9406	0.9535	1.2567	9.6203	1.9102
2	0.9403	0.9387	0.9463	1.4047	11.015	2.0522
3	0.9373	0.9357	0.9407	1.4454	11.0156	2.1577
4	0.8535	0.48496	0.8548	2.3919	17.2805	3.3754
5	0.8554	0.8515	0.8538	2.3743	16.9669	3.3871
1 (feat. select)	0.9417	0.9408	0.9536	1.2595	9.6371	1.9075
5 (feat.	0.8536	0.8514	0.8525	2.39	17.0933	3.402

select)						
---------	--	--	--	--	--	--

Model No.	Normality	Constant Variance	Equal Mean	Independence	Linear
0	No	No	Yes	Yes	No
1	Yes	No	Yes	Yes	No
2	Yes	No	Yes	Yes	No
3	Yes	No	Yes	Yes	No
4	Yes	No	Yes	Yes	Yes
5	Yes	Yes	Yes	Yes	Yes
1 (feat. select)	Yes	No	Yes	Yes	No
5 (feat. select)	Yes	Yes	Yes	Yes	Yes

Metric Analysis for the Decision Tree:

Confusion Matrix and Statistics

		Reference						
Prediction		1	1.25	1.5	1.75	2	2.5	3
1	185036	3259	1396	624	457	40	2	
1.25	0	0	0	0	0	0	0	
1.5	0	0	0	0	0	0	0	
1.75	0	0	0	0	0	0	0	
2	64	66	123	102	214	6	1	
2.5	0	0	0	0	0	0	0	
3	0	0	0	0	0	0	0	

Overall Statistics

Accuracy : 0.9679
 95% CI : (0.9671, 0.9687)
 No Information Rate : 0.9671
 P-Value [Acc > NIR] : 0.02729

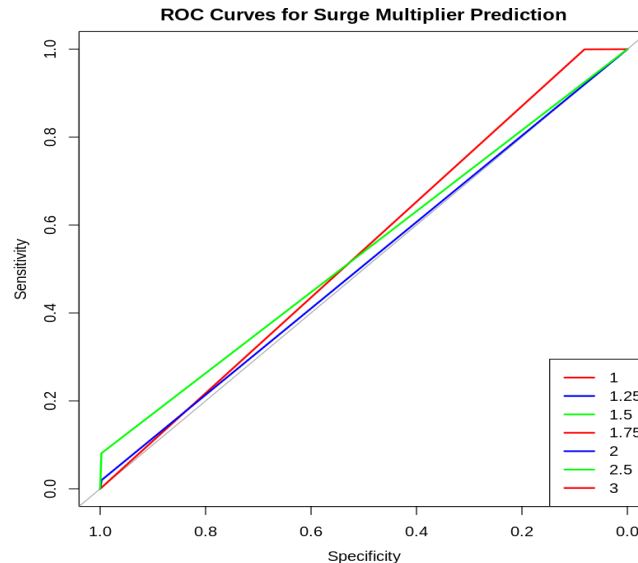
 Kappa : 0.103

The metric analysis for the surge multiplier model highlights some significant challenges in how the data is distributed and how well the model is performing. The dataset contains over 693,000 rows, but a vast majority of the data points are classified as having a surge multiplier of 1, creating a noticeable

imbalance. This becomes evident in the confusion matrix, where the model heavily predicts the surge multiplier as 1, almost ignoring other values like 1.25, 1.5, 1.75, 2, 2.5, and 3.

When we look closer, the model's sensitivity, its ability to correctly identify each class, is almost perfect for 1 at 99.97%, but it completely misses the mark for other surge multipliers, with zero detection for 1.25, 1.5, 1.75, 2.5, and 3. It only manages to detect 31.89% of the cases for a surge multiplier of 2, indicating that the model is heavily biased toward predicting the most common class while neglecting the rarer, yet potentially more significant surge levels.

The decision tree visualization drives this point home. The primary and only significant split in the tree is based on the price, with a threshold set at 52. If the price is less than 52, the model almost always predicts the surge multiplier as 1, with a probability of 97%. On the other hand, if the price exceeds 52, the model slightly opens up to predict higher surge multipliers, but even then, the predictions remain heavily skewed.



The ROC curves reinforce this imbalance, showing that the model is performing well for class 1, with a curve that almost perfectly aligns with the upper boundary. However, for other classes, the ROC curves barely lift off the baseline, indicating that the model is essentially guessing for these classes.

Despite the seemingly high accuracy of 96.79%, this result is misleading because it is driven by the model's over-reliance on predicting the dominant class 1. This imbalance masks the model's inability to effectively identify and classify other surge multipliers. To address this, we could explore methods like oversampling, undersampling, or introducing class weights to balance the training data and allow the model to learn more effectively from the less common surge multiplier values.

Precision and F1 Score Analysis:

The precision and F1 Score analysis for the surge multiplier prediction model offers a deeper look into how well the model handles class imbalance in the dataset. For the dominant class 1, the precision was calculated as approximately 0.9697, meaning that when the model predicted 1, it was correct almost 97% of the time. The recall for class 1 was nearly perfect at 0.9997, indicating that the model successfully captured nearly all actual 1 instances. However, the story was different for less common classes like 1.5 and 2. Here, both precision and recall dropped significantly, leading to F1 Scores close to zero. This imbalance highlights how the model is strongly biased towards the dominant 1 class, often at the expense of minority classes.

The F1 Score, which balances precision and recall, was particularly insightful. For the dominant class 1, the F1 Score was 0.9844, reflecting both high precision and recall. However, the F1 Scores for other classes remained alarmingly low, indicating the model's struggle to effectively capture the less frequent surge multiplier values. This discrepancy underscores the need for more sophisticated techniques, like resampling or adjusting class weights, to handle the imbalance more effectively.

Additionally, the ROC curves and AUC values were calculated for each surge multiplier class, providing a visual representation of how well the model distinguishes between classes. The dominant class 1 had a relatively strong AUC, suggesting effective discrimination between true positives and false positives. In contrast, minority classes showed lower AUC values, further confirming the model's difficulty in accurately identifying these less common surge multiplier values. While the model performs exceptionally well for the majority class 1, its effectiveness significantly declines for less frequent classes like 1.5, 2, and beyond. This imbalance skews overall performance metrics, making precision and F1 Score more informative than accuracy in evaluating the model. The analysis points to the need for targeted optimization strategies, such as resampling or applying cost-sensitive learning, to improve predictive power for underrepresented classes without sacrificing the model's strong performance for the dominant class.

Validation of Logistic Regression:**Model 1: Cab Type Prediction**

The validation results for Model 1, which predicts the cab_type (Uber or Lyft), indicate that the model performs only slightly better than random guessing. With an overall accuracy of 54.17%, the model correctly predicted 243,019 Uber rides and 102,544 Lyft rides, but it also misclassified a substantial number of 204,864 Lyft rides as Uber. This imbalance in predictions suggests the model has a tendency to favor Uber over Lyft, which could be due to patterns in the training data or an underlying class imbalance. Although the model captures some of the structure in the data, its low accuracy and strong bias make it unreliable for practical use without further refinement.

Confusion Matrix:

	Actual	
Predicted	Lyft	Uber
FALSE	102544	87549
TRUE	204864	243019

Model Accuracy: 0.5416552 --> 54.17 %

Model 2: Predicting Surge Pricing (is_surge)

The validation results for the is_surge model, which predicts whether a ride is under surge pricing, show a high overall accuracy of 96.71%. The model correctly classified the vast majority of non-surge rides (617,001 out of 637,976 total) and falsely labeled only 20,975 surge rides as non-surge. This high accuracy suggests that the model is highly effective at identifying non-surge rides, but it may underpredict surge events. The skewed predictions could be due to class imbalance, where non-surge rides are far more common in the dataset. Despite this, the model demonstrates strong overall performance for detecting surge pricing patterns.

Confusion Matrix:

	Actual	
Predicted	0	1
FALSE	617001	20975

Model Accuracy: 0.9671226 --> 96.71 %

Section 5: Discussion & Conclusion

Main Conclusion

Overall, the project's goal was successful for Regression. We managed to explore the data and garner hidden insight into the dataset. Moreover, we were able to set up several models based on different remedial measures, which were based on larger problems. We managed to have 5 models running, each slowly incremented, eventually leading us to model #5, which was able to fix the 5 assumptions of linear transformations. In the end, we found that a combination of outlier removal, box-cox, WLS, and polynomial regression made this the best algorithm against the 5 assumptions. Moreover, we found other models that were able to garner large values of R^2 .

Challenges & Future Works

Several challenges were present to us while working on the data. For one, we had a limited time to perform both sequential EDA, followed by regression, hence we had to perform both task simultaneously. Moreover, the dataset itself possessed several problems. The data had not been as uniform as thought about earlier, with several days & hours missing between the range. Moreover, the data itself, though large, its roughly less than 2 months, as such, it in of itself is a type of sample. That cannot be extended to statistical read about the whole population (i.e. the rest of the time in the world).

Key days are missing, especially like christmas, which may end up making day or significant. The data might not have been sythensized but the pre-processing of the data done on kaggle by the data collection method has a similar affect, making it more biased, and unreliable (i.e. rounding certain values). For future works, we saw that WLS, with box-cox and polynomial regression was our strongest tool for performance. Hence, we would like to explore those perspective more, as throughout the course, we focused on linear regression. Moreover, we would like to work with a more uniform representation of the data, because while it does have potential, the gaps do make some sort of hidden pattern that becomes difficult to work with.

Section 7: Appendix & R-Code

Appendix A: Kaggle Rideshare Dataset Feature Description

Feature	Data Type	Description
id	character	This is a unique identifier assigned to each ride entry.
timestamp	numeric	This shows the exact time the ride was requested.
hour	integer	This indicates the hour of the day when the ride was requested.
day	integer	This records the day of the month the ride occurred.
month	integer	This captures the month in which the ride took place.
datetime	character	This represents the full timestamp (date and time) of the ride.
timezone	character	This shows the timezone relevant to the ride's location.
source	character	This is the starting point or pickup location of the ride.
destination	character	This is the endpoint or drop-off location of the ride.
cab_type	character	This tells whether the ride was booked through Uber or Lyft.
product_id	character	This is the internal ID used by the company to identify a specific ride option.
name	character	This refers to the specific ride type like UberX, Lyft Lux, etc.
price	numeric	This indicates the estimated or actual fare for the ride.
distance	numeric	This shows how far the ride traveled, measured in miles.
surge_multiplier	numeric	This represents how much the base fare was multiplied due to surge pricing.

latitude	numeric	This gives the latitude coordinate of the ride's pickup location.
longitude	numeric	This gives the longitude coordinate of the ride's pickup location.
temperature	numeric	This records the actual temperature at the time of the ride.
apparentTemperature	numeric	This is the 'feels like' temperature at the time of the ride.
short_summary	character	This provides a short weather description, like 'Rain' or 'Clear'.
long_summary	character	This provides a more detailed weather summary.
precipIntensity	numeric	This shows the intensity of precipitation, like rain or snow.
precipProbability	numeric	This is the probability of precipitation occurring during the ride.
humidity	numeric	This shows the humidity level as a percentage.
windSpeed	numeric	This records the wind speed at the time of the ride in mph.
windGust	numeric	This shows the highest wind gust speed recorded.
windGustTime	integer	This indicates the time when the highest wind gust occurred.
visibility	numeric	This shows how clear the visibility was during the ride, measured in miles.
temperatureHigh	numeric	This is the highest temperature recorded on that day.
temperatureHighTime	integer	This tells the time when the highest temperature occurred.
temperatureLow	numeric	This is the lowest temperature recorded on that day.

temperatureLowTime	integer	This tells the time when the lowest temperature occurred.
apparentTemperatureHigh	numeric	This is the highest 'feels like' temperature on that day.
apparentTemperatureHighTime	integer	This is the time that the highest 'feels like' temperature was recorded.
apparentTemperatureLow	numeric	This is the lowest 'feels like' temperature on that day.
apparentTemperatureLowTime	integer	This is the time that the lowest 'feels like' temperature was recorded.
icon	character	This is a visual weather icon label like 'rain' or 'clear-day'.
dewPoint	numeric	This records the temperature at which dew forms, indicating humidity.
pressure	numeric	This shows the atmospheric pressure in millibars.
windBearing	integer	This indicates the direction the wind is coming from in degrees.
cloudCover	numeric	This represents the percentage of the sky covered by clouds.
uvIndex	integer	This measures the intensity of ultraviolet radiation at that time.
visibility.1	numeric	This is a duplicate of visibility, possibly due to merging datasets.
ozone	numeric	This indicates the ozone concentration in the atmosphere.
sunriseTime	integer	This records the time of sunrise for that day.
sunsetTime	integer	This records the time of sunset for that day.

moonPhase	numeric	This represents the phase of the moon, scaled from 0 (new moon) to 1 (next new moon).
precipIntensityMax	numeric	This is the maximum intensity of precipitation for that day.
uvIndexTime	integer	This is the time of day when the UV index reached its maximum.
temperatureMin	numeric	This is the minimum actual temperature for that day.
temperatureMinTime	integer	This shows the time when the minimum temperature occurred.
temperatureMax	numeric	This is the maximum actual temperature for that day.
temperatureMaxTime	integer	This shows the time when the maximum temperature occurred.
apparentTemperatureMin	numeric	This is the lowest 'feels like' temperature of the day.
apparentTemperatureMinTime	integer	This is the time when the lowest 'feels like' temperature occurred.
apparentTemperatureMax	numeric	This is the highest 'feels like' temperature of the day.
apparentTemperatureMaxTime	integer	This is the time when the highest 'feels like' temperature occurred.

Code Available in the Code File (Too Large to Attach Feasible to the Report)
Along with the Output